Solve Lambda Vacuum Field Equations

Case 6D_1A : 4D spacetime (t,r,θ,ϕ) + two individually curled axis (ψ_4,ψ_5)
1: Curled geometries: Circles
A: Curled metric - a[t]^2 $\mathcal{D}_4$[r] $p_4{}^2$ d$\psi_4{}^2$ – a[t]^2 $\mathcal{D}_5$[r] $p_5{}^2$ d$\psi_5{}^2$
3-space metric: -B(t,r) dr^2 – $\mathbb{C}$(t,r) r^2 dθ^2 – $\mathbb{C}$(t,r) r^2 sin$^2\theta$ dϕ^2
Time metric: A(t,r) dt^2

Line Element

Out[ ]=

$$ds^2 = A\ dt^2 - B\ dr^2 - r^2\,\mathbb{C}\ d\theta^2 - r^2\,\mathbb{C}\,\mathrm{Sin}[\theta]^2\ d\phi^2 - a^2\,p_4^2\,\mathcal{D}_4\ d\psi_4^2 - a^2\,p_5^2\,\mathcal{D}_5\ d\psi_5^2$$

Metric Tensor Matrices

Out[ ]=

$$g:\ (g_{\mu\nu}) = \begin{pmatrix} \frac{A}{c^2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -B & 0 & 0 & 0 & 0 \\ 0 & 0 & -r^2\,\mathbb{C} & 0 & 0 & 0 \\ 0 & 0 & 0 & -r^2\,\mathbb{C}\,\mathrm{Sin}[\theta]^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & -a^2\,p_4^2\,\mathcal{D}_4 & 0 \\ 0 & 0 & 0 & 0 & 0 & -a^2\,p_5^2\,\mathcal{D}_5 \end{pmatrix}$$

$$G:\ (g^{\mu\nu}) = \begin{pmatrix} \frac{c^2}{A} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{B} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{r^2\,\mathbb{C}} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\mathrm{Csc}[\theta]^2}{r^2\,\mathbb{C}} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{a^2\,p_4^2\,\mathcal{D}_4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{a^2\,p_5^2\,\mathcal{D}_5} \end{pmatrix}$$

Case 6D_1A: 6D = 4D spacetime (t,r,θ,ϕ) + two individually curled axis (ψ_4,ψ_5)
$g_{00}=\frac{A(t,r)}{c^2}$, $g_{11}=-B(t,r)$, $g_{22}=-\mathbb{C}(t,r)r^2$, $g_{33}=-\mathbb{C}(t,r)r^2\sin^2\theta$
$g_{44}=-a(t)^2\mathcal{D}_4(r)p_4{}^2$, $g_{55}=-a(t)^2\mathcal{D}_5(r)p_5{}^2$. a(t), $\mathcal{D}_4$(r) & $\mathcal{D}_5$(r) not constant

We wish to be explicit up-front that in Case 1A **we assume $\mathcal{D}_4\neq\mathcal{D}_5$** because $\mathcal{D}_4=\mathcal{D}_5$ is handled in Case 1E. Thus, as we solve equations, we do not consider setting $\mathcal{D}_4=\mathcal{D}_5$ even if it might be helpful.

The Lambda field equations can be represented as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$:

$$R_{00}: \quad \frac{2 c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{a A^2} - \frac{c^2 \dot{A} \dot{B}}{4 A^2 B} - \frac{c^2 \dot{B}^2}{4 A B^2} + \frac{c^2 \ddot{B}}{2 A B} - \frac{c^2 \dot{A} \dot{\mathbb{C}}}{2 A^2 \mathbb{C}} - \frac{c^2 \dot{\mathbb{C}}^2}{2 A \mathbb{C}^2} + \frac{c^2 \ddot{\mathbb{C}}}{A \mathbb{C}} - \frac{A'}{A B r} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' \mathbb{C}'}{2 A B \mathbb{C}} - \frac{A' \mathbb{D}_4'}{4 A B \mathbb{D}_4} - \frac{A' \mathbb{D}_5'}{4 A B \mathbb{D}_5} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{B}}{B r} + \frac{\dot{\mathbb{C}}}{r \mathbb{C}} - \frac{\dot{a} A'}{a A} - \frac{\dot{\mathbb{C}} A'}{2 A \mathbb{C}} - \frac{\dot{B} \mathbb{C}'}{2 B \mathbb{C}} - \frac{\dot{\mathbb{C}} \mathbb{C}'}{2 \mathbb{C}^2} + \frac{\dot{\mathbb{C}}'}{\mathbb{C}} + \frac{\dot{a} \mathbb{D}_4'}{2 a \mathbb{D}_4} - \frac{\dot{B} \mathbb{D}_4'}{4 B \mathbb{D}_4} + \frac{\dot{a} \mathbb{D}_5'}{2 a \mathbb{D}_5} - \frac{\dot{B} \mathbb{D}_5'}{4 B \mathbb{D}_5} = 0$$

$$R_{11}: \quad \frac{c^2 \dot{a} \dot{B}}{a A B} - \frac{c^2 \dot{A} \dot{B}}{4 A^2 B} - \frac{c^2 \dot{B}^2}{4 A B^2} + \frac{c^2 \ddot{B}}{2 A B} + \frac{c^2 \dot{B} \dot{\mathbb{C}}}{2 A B \mathbb{C}} + \frac{(A')^2}{4 A^2 B} + \frac{B'}{B^2 r} + \frac{A' B'}{4 A B^2} - \frac{2 \mathbb{C}'}{B r \mathbb{C}} + \frac{B' \mathbb{C}'}{2 B^2 \mathbb{C}} + \frac{(\mathbb{C}')^2}{2 B \mathbb{C}^2} + \frac{B' \mathbb{D}_4'}{4 B^2 \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 B \mathbb{D}_4^2} + \frac{B' \mathbb{D}_5'}{4 B^2 \mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4 B \mathbb{D}_5^2} - \frac{A''}{2 A B} - \frac{\mathbb{C}''}{B \mathbb{C}} - \frac{\mathbb{D}_4''}{2 B \mathbb{D}_4} - \frac{\mathbb{D}_5''}{2 B \mathbb{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{1}{B r^2} + \frac{1}{r^2 \mathbb{C}} + \frac{c^2 \dot{a} \dot{\mathbb{C}}}{a A \mathbb{C}} - \frac{c^2 \dot{A} \dot{\mathbb{C}}}{4 A^2 \mathbb{C}} + \frac{c^2 \dot{B} \dot{\mathbb{C}}}{4 A B \mathbb{C}} + \frac{c^2 \ddot{\mathbb{C}}}{2 A \mathbb{C}} - \frac{A'}{2 A B r} + \frac{B'}{2 B^2 r} - \frac{2 \mathbb{C}'}{B r \mathbb{C}} - \frac{A' \mathbb{C}'}{4 A B \mathbb{C}} + \frac{B' \mathbb{C}'}{4 B^2 \mathbb{C}} - \frac{\mathbb{D}_4'}{2 B r \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{4 B \mathbb{C} \mathbb{D}_4} - \frac{\mathbb{D}_5'}{2 B r \mathbb{D}_5} - \frac{\mathbb{C}' \mathbb{D}_5'}{4 B \mathbb{C} \mathbb{D}_5} - \frac{\mathbb{C}''}{2 B \mathbb{C}} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{c^2 \dot{a} \dot{B}}{2 a A B} + \frac{c^2 \dot{a} \dot{\mathbb{C}}}{a A \mathbb{C}} - \frac{\mathbb{D}_4'}{B r \mathbb{D}_4} - \frac{A' \mathbb{D}_4'}{4 A B \mathbb{D}_4} + \frac{B' \mathbb{D}_4'}{4 B^2 \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{2 B \mathbb{C} \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 B \mathbb{D}_4^2} - \frac{\mathbb{D}_4' \mathbb{D}_5'}{4 B \mathbb{D}_4 \mathbb{D}_5} - \frac{\mathbb{D}_4''}{2 B \mathbb{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{c^2 \dot{a} \dot{B}}{2 a A B} + \frac{c^2 \dot{a} \dot{\mathbb{C}}}{a A \mathbb{C}} - \frac{\mathbb{D}_5'}{B r \mathbb{D}_5} - \frac{A' \mathbb{D}_5'}{4 A B \mathbb{D}_5} + \frac{B' \mathbb{D}_5'}{4 B^2 \mathbb{D}_5} - \frac{\mathbb{C}' \mathbb{D}_5'}{2 B \mathbb{C} \mathbb{D}_5} - \frac{\mathbb{D}_4' \mathbb{D}_5'}{4 B \mathbb{D}_4 \mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4 B \mathbb{D}_5^2} - \frac{\mathbb{D}_5''}{2 B \mathbb{D}_5} = \Lambda$$

We observed in Notebook 6D_1 that R_{33} is redundant.
Hence, R_{33} is omitted here and in all of the 6D notebooks.

The remaining equations are too
complex to solve without the usual additional assumptions.

a: $B(t,r) = a(t)^2 B(r)$, $\mathbb{C}(t,r) = a(t)^2 C(r)$

The field equations become

R_{00}:

$$\frac{5 c^2 \ddot{a}}{a A} - \frac{5 c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{a^2 A r B} + \frac{(A')^2}{4 a^2 A^2 B} - \frac{A' C'}{2 a^2 A C B} + \frac{A' B'}{4 a^2 A B^2} - \frac{A' \mathbb{D}_4'}{4 a^2 A B \mathbb{D}_4} - \frac{A' \mathbb{D}_5'}{4 a^2 A B \mathbb{D}_5} - \frac{A''}{2 a^2 A B} = \Lambda$$

$$R_{01}: \quad -\frac{2 \dot{a} A'}{a A} = 0$$

$$R_{11}: \quad \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2 B} - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{A' B'}{4 a^2 A B^2} + \frac{C' B'}{2 a^2 C B^2} + \frac{B' \mathbb{D}_4'}{4 a^2 B^2 \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 a^2 B \mathbb{D}_4^2} + \frac{B' \mathbb{D}_5'}{4 a^2 B^2 \mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4 a^2 B \mathbb{D}_5^2} - \frac{A''}{2 a^2 A B} - \frac{C''}{a^2 C B} - \frac{\mathbb{D}_4''}{2 a^2 B \mathbb{D}_4} - \frac{\mathbb{D}_5''}{2 a^2 B \mathbb{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{2 a^2 A r B} - \frac{2 C'}{a^2 r C B} - \frac{A' C'}{4 a^2 A C B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{D_4'}{2 a^2 r B D_4} - \frac{C' D_4'}{4 a^2 C B D_4} - \frac{D_5'}{2 a^2 r B D_5} - \frac{C' D_5'}{4 a^2 C B D_5} - \frac{C''}{2 a^2 C B} = \Lambda$$

$$R_{44}: \quad \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{D_4'}{a^2 r B D_4} - \frac{A' D_4'}{4 a^2 A B D_4} - \frac{C' D_4'}{2 a^2 C B D_4} + \frac{B' D_4'}{4 a^2 B^2 D_4} + \frac{(D_4')^2}{4 a^2 B D_4^2} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} - \frac{D_4''}{2 a^2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{D_5'}{a^2 r B D_5} - \frac{A' D_5'}{4 a^2 A B D_5} - \frac{C' D_5'}{2 a^2 C B D_5} + \frac{B' D_5'}{4 a^2 B^2 D_5} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} + \frac{(D_5')^2}{4 a^2 B D_5^2} - \frac{D_5''}{2 a^2 B D_5} = \Lambda$$

Since $a(t)$ is not allowed to be a constant in Case a, then $\dot{a} \neq 0$, and from R_{01} we get that A is a function solely of t:

$$A(t,r) = \mathcal{A}(t)$$

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{5 c^2 \ddot{a}}{a \mathcal{A}} - \frac{5 c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{4 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C' B'}{2 a^2 C B^2} + \frac{B' D_4'}{4 a^2 B^2 D_4} + \frac{(D_4')^2}{4 a^2 B D_4^2} + \frac{B' D_5'}{4 a^2 B^2 D_5} + \frac{(D_5')^2}{4 a^2 B D_5^2} - \frac{C''}{a^2 C B} - \frac{D_4''}{2 a^2 B D_4} - \frac{D_5''}{2 a^2 B D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \frac{4 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{2 C'}{a^2 r C B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{D_4'}{2 a^2 r B D_4} - \frac{C' D_4'}{4 a^2 C B D_4} - \frac{D_5'}{2 a^2 r B D_5} - \frac{C' D_5'}{4 a^2 C B D_5} - \frac{C''}{2 a^2 C B} = \Lambda$$

$$R_{44}: \quad \frac{4 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2 a^2 C B D_4} + \frac{B' D_4'}{4 a^2 B^2 D_4} + \frac{(D_4')^2}{4 a^2 B D_4^2} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} - \frac{D_4''}{2 a^2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{4 c^2 \dot{a}^2}{a^2 \mathcal{A}} + \frac{c^2 \ddot{a}}{a \mathcal{A}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{D_5'}{a^2 r B D_5} - \frac{C' D_5'}{2 a^2 C B D_5} + \frac{B' D_5'}{4 a^2 B^2 D_5} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} + \frac{(D_5')^2}{4 a^2 B D_5^2} - \frac{D_5''}{2 a^2 B D_5} = \Lambda$$

We can solve R_{00} for $\mathcal{A}(t)$:

$$\mathcal{A}(t) = \frac{5 c^2 \dot{a}^2}{5 c^2 k + a^2 \Lambda}$$ where k is a constant of integration.

Plugging this gives us 4 equations in the 4 unknowns $B(r)$, $C(r)$, $D_4(r)$ and $D_5(r)$:

$$R_{11}: \quad \frac{4 c^2 k}{a^2} + \Lambda - \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{B'}{a^2 r B^2} + \frac{C' B'}{2 a^2 C B^2} + \frac{B' D_4'}{4 a^2 B^2 D_4} + \frac{(D_4')^2}{4 a^2 B D_4^2} + \frac{B' D_5'}{4 a^2 B^2 D_5} + \frac{(D_5')^2}{4 a^2 B D_5^2} - \frac{C''}{a^2 C B} - \frac{D_4''}{2 a^2 B D_4} - \frac{D_5''}{2 a^2 B D_5} = \Lambda$$

$$R_{22}: \quad \frac{4 c^2 k}{a^2} + \frac{1}{a^2 r^2 C} - \frac{1}{a^2 r^2 B} + \Lambda - \frac{2 C'}{a^2 r C B} + \frac{B'}{2 a^2 r B^2} + \frac{C' B'}{4 a^2 C B^2} - \frac{D_4'}{2 a^2 r B D_4} - \frac{C' D_4'}{4 a^2 C B D_4} - \frac{D_5'}{2 a^2 r B D_5} - \frac{C' D_5'}{4 a^2 C B D_5} - \frac{C''}{2 a^2 C B} = \Lambda$$

$$R_{44}: \quad \frac{4 c^2 k}{a^2} + \Lambda - \frac{D_4'}{a^2 r B D_4} - \frac{C' D_4'}{2 a^2 C B D_4} + \frac{B' D_4'}{4 a^2 B^2 D_4} + \frac{(D_4')^2}{4 a^2 B D_4^2} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} - \frac{D_4''}{2 a^2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{4 c^2 k}{a^2} + \Lambda - \frac{D_5'}{a^2 r B D_5} - \frac{C' D_5'}{2 a^2 C B D_5} + \frac{B' D_5'}{4 a^2 B^2 D_5} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} + \frac{(D_5')^2}{4 a^2 B D_5^2} - \frac{D_5''}{2 a^2 B D_5} = \Lambda$$

These equations, individually, are too difficult to be solved. However, $R_{11}-2R_{22}-R_{44}-R_{55}$ can be algebraically solved for B or Dsolved for any of C, D_4, or D_5.

The algebraic formula for $B(r)$ is somewhat complex and makes all the equations lengthy and unwieldy when it is plugged back in. So, we investigate solving for D_5.

$$R_{11}-2R_{22}-R_{44}-R_{55}: \quad -\frac{12 c^2 k}{a^2} - \frac{2}{a^2 r^2 C} + \frac{2}{a^2 r^2 B} - 3 \Lambda + \frac{2 C'}{a^2 r C B} + \frac{(C')^2}{2 a^2 C^2 B} + \frac{2 D_4'}{a^2 r B D_4} + \frac{C' D_4'}{a^2 C B D_4} + \frac{2 D_5'}{a^2 r B D_5} + \frac{C' D_5'}{a^2 C B D_5} + \frac{D_4' D_5'}{2 a^2 B D_4 D_5} = -3 \Lambda$$

$$D_5(r) = e^{\int_1^r \frac{-\rho^2 D_4 (C')^2 + 4 C^2 \left(\left(-1+6 c^2 k B \rho^2\right) D_4 - \rho D_4'\right) + C \left(4 B D_4 - 2 \rho C' \left(2 D_4 + \rho D_4'\right)\right)}{C \rho \left(2 \rho D_4 C' + C \left(4 D_4 + \rho D_4'\right)\right)} d\rho} \alpha$$ where α is a constant of integration.

Plugging back in, the integral conveniently disappears from all of the equations, but they are nonetheless quite lengthy! However, $R_{11}-R_{55}$ is of reasonable length, and since it has zero on the RHS, the LHS can be put over a common denominator and then the denominator ignored:

$$R_{11}-R_{55}: \quad r^2 B D_4^2 (C')^2 + 2 C D_4 \left(2 B^2 D_4 + r^2 D_4 C' B' - r B \left(r C' D_4' + 2 D_4 \left(3 C' + r C''\right)\right)\right) + C^2 \left(24 c^2 k r^2 B^2 D_4^2 + r D_4 B' \left(4 D_4 + r D_4'\right) + B \left(-4 D_4^2 + r^2 (D_4')^2 - 2 r D_4 \left(2 D_4' + r D_4''\right)\right)\right) = 0$$

Of the 3 unknowns, only $B(r)$ admits a solution, and it is a complex formula that has an integral in both its numerator and denominator. The resulting equations and combinations of equations are all unwieldy.

This case is too difficult to solve.

The normal next steps are to explore 3 subcases of Case a:

(a_II) A(t,r)=c²

(ai) B(r)=1 [i.e., $B(t,r)=a(t)^2$]

(aii) C(r)=1 [i.e., $\mathbb{C}(t,r)=a(t)^2$]

a_II: $B(t,r)=a(t)^2 B(r)$, $\mathbb{C}(t,r)=a(t)^2 C(r)$, $A(t,r)=c^2$

Re-doing Case a from scratch, the field equations become

$$R_{00}: \quad \frac{5\,\ddot{a}}{a} = \Lambda$$

$$R_{11}: \quad \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{C'\,B'}{2\,a^2\,C\,B^2} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{C''}{a^2\,C\,B} - \frac{D_4''}{2\,a^2\,B\,D_4} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2\,C} - \frac{1}{a^2\,r^2\,B} + \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,C'}{a^2\,r\,C\,B} + \frac{B'}{2\,a^2\,r\,B^2} + \frac{C'\,B'}{4\,a^2\,C\,B^2} - \frac{D_4'}{2\,a^2\,r\,B\,D_4} - \frac{C'\,D_4'}{4\,a^2\,C\,B\,D_4} - \frac{D_5'}{2\,a^2\,r\,B\,D_5} - \frac{C'\,D_5'}{4\,a^2\,C\,B\,D_5} - \frac{C''}{2\,a^2\,C\,B} = \Lambda$$

$$R_{44}: \quad \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_4'}{a^2\,r\,B\,D_4} - \frac{C'\,D_4'}{2\,a^2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad \frac{4\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_5'}{a^2\,r\,B\,D_5} - \frac{C'\,D_5'}{2\,a^2\,C\,B\,D_5} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

Solving R_{00} yields

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{5}}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{5}}}\,\beta$$ where α and β are constants of integration.

We can also algebraically solve $R_{11}-2R_{22}-R_{44}-R_{55}$ for $B(r)$:

$$R_{11}-2R_{22}-R_{44}-R_{55}: \quad -\frac{2}{a^2\,r^2\,C} + \frac{2}{a^2\,r^2\,B} - \frac{12\,\dot{a}^2}{a^2} - \frac{3\,\ddot{a}}{a} + \frac{2\,C'}{a^2\,r\,C\,B} + \frac{(C')^2}{2\,a^2\,C^2\,B} + \frac{2\,D_4'}{a^2\,r\,B\,D_4} + \frac{C'\,D_4'}{a^2\,C\,B\,D_4} + \frac{2\,D_5'}{a^2\,r\,B\,D_5} + \frac{C'\,D_5'}{a^2\,C\,B\,D_5} + \frac{D_4'\,D_5'}{2\,a^2\,B\,D_4\,D_5} = -3\,\Lambda$$

$$B(r) = \left(\left(-4\,C^2\,D_4\,D_5 - 4\,r\,C\,D_4\,D_5\,C' - r^2\,D_4\,D_5\,(C')^2 - 4\,r\,C^2\,D_5\,D_4' - 2\,r^2\,C\,D_5\,C'\,D_4' - 4\,r\,C^2\,D_4\,D_5' - 2\,r^2\,C\,D_4\,C'\,D_5' - r^2\,C^2\,D_4'\,D_5'\right) \Big/ \left(2\,C\,\left(-2 + 3\,a^2\,r^2\,C\,\Lambda - 12\,r^2\,C\,\dot{a}^2 - 3\,a\,r^2\,C\,\ddot{a}\right)\,D_4\,D_5\right)\right)$$

In order for $B(r)$ to strictly be a function of r, the t terms in the denominator must disappear. There are no terms in the numerator to cancel them out, so the denominator t-terms must be self-canceling. It turns out they are. Call the denominator expression in the parentheses $p(t,r)$:

$$p(t,r) = -2 + 3\,a^2\,r^2\,C\,\Lambda - 12\,r^2\,C\,\dot{a}^2 - 3\,a\,r^2\,C\,\ddot{a}$$

When we plug in the formula above for a(t), this simplifies to

$$p(t,r) = -2 + \frac{48}{5} r^2 \alpha \beta \Lambda \mathbb{C}[r]$$ and the t-terms have disappeared

$$\Rightarrow \quad B(r) = \left(\left(-4 \mathbb{C}^2 \mathbb{D}_4 \mathbb{D}_5 - 4 r \mathbb{C} \mathbb{D}_4 \mathbb{D}_5 \mathbb{C}' - r^2 \mathbb{D}_4 \mathbb{D}_5 (\mathbb{C}')^2 - 4 r \mathbb{C}^2 \mathbb{D}_5 \mathbb{D}_4' - 2 r^2 \mathbb{C} \mathbb{D}_5 \mathbb{C}' \mathbb{D}_4' - 4 r \mathbb{C}^2 \mathbb{D}_4 \mathbb{D}_5' - 2 r^2 \mathbb{C} \mathbb{D}_4 \mathbb{C}' \mathbb{D}_5' - r^2 \mathbb{C}^2 \mathbb{D}_4' \mathbb{D}_5'\right) \Big/ \left(2 \mathbb{C} \left(-2 + \frac{48}{5} r^2 \mathbb{C} \alpha \beta \Lambda\right) \mathbb{D}_4 \mathbb{D}_5\right)\right),$$ strictly a function of r.

When we plug $\mathbb{B}(r)$ into the remaining equations, they are each pages long and I can't find an algebraic combination that simplifies them. Similar to the 5D case, the terms on the LHS of R_{11}-R_{44}-R_{55} all have the same denominator, but this time the RHS is $-\Lambda$, not 0. So we cannot simply ignore the denominators. Even after setting β=0, these equations are too lengthy to display here, and certainly too difficult for me to solve.

ai: $B(t,r)=a[t]^2$, $\mathbb{C}(t,r)=a(t)^2\mathbb{C}(r)$

Starting fresh, the field equations become

$$R_{00}: \quad \frac{5 c^2 \ddot{a}}{a A} - \frac{5 c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{a^2 A r} + \frac{(A')^2}{4 a^2 A^2} - \frac{A' \mathbb{C}'}{2 a^2 A \mathbb{C}} - \frac{A' \mathbb{D}_4'}{4 a^2 A \mathbb{D}_4} - \frac{A' \mathbb{D}_5'}{4 a^2 A \mathbb{D}_5} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{01}: \quad -\frac{2 \dot{a} A'}{a A} = 0$$

$$R_{11}: \quad \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} + \frac{(A')^2}{4 a^2 A^2} - \frac{2 \mathbb{C}'}{a^2 r \mathbb{C}} + \frac{(\mathbb{C}')^2}{2 a^2 \mathbb{C}^2} + \frac{(\mathbb{D}_4')^2}{4 a^2 \mathbb{D}_4^2} + \frac{(\mathbb{D}_5')^2}{4 a^2 \mathbb{D}_5^2} - \frac{A''}{2 a^2 A} - \frac{\mathbb{C}''}{a^2 \mathbb{C}} - \frac{\mathbb{D}_4''}{2 a^2 \mathbb{D}_4} - \frac{\mathbb{D}_5''}{2 a^2 \mathbb{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2 r^2} + \frac{1}{a^2 r^2 \mathbb{C}} + \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{2 a^2 A r} - \frac{2 \mathbb{C}'}{a^2 r \mathbb{C}} - \frac{A' \mathbb{C}'}{4 a^2 A \mathbb{C}} - \frac{\mathbb{D}_4'}{2 a^2 r \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{4 a^2 \mathbb{C} \mathbb{D}_4} - \frac{\mathbb{D}_5'}{2 a^2 r \mathbb{D}_5} - \frac{\mathbb{C}' \mathbb{D}_5'}{4 a^2 \mathbb{C} \mathbb{D}_5} - \frac{\mathbb{C}''}{2 a^2 \mathbb{C}} = \Lambda$$

$$R_{44}: \quad \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{\mathbb{D}_4'}{a^2 r \mathbb{D}_4} - \frac{A' \mathbb{D}_4'}{4 a^2 A \mathbb{D}_4} - \frac{\mathbb{C}' \mathbb{D}_4'}{2 a^2 \mathbb{C} \mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4 a^2 \mathbb{D}_4^2} - \frac{\mathbb{D}_4' \mathbb{D}_5'}{4 a^2 \mathbb{D}_4 \mathbb{D}_5} - \frac{\mathbb{D}_4''}{2 a^2 \mathbb{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{4 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{\mathbb{D}_5'}{a^2 r \mathbb{D}_5} - \frac{A' \mathbb{D}_5'}{4 a^2 A \mathbb{D}_5} - \frac{\mathbb{C}' \mathbb{D}_5'}{2 a^2 \mathbb{C} \mathbb{D}_5} - \frac{\mathbb{D}_4' \mathbb{D}_5'}{4 a^2 \mathbb{D}_4 \mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4 a^2 \mathbb{D}_5^2} - \frac{\mathbb{D}_5''}{2 a^2 \mathbb{D}_5} = \Lambda$$

From R_{01}, we again get that

$$A(t,r) = \mathcal{A}(t)$$

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{5\,c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{5\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} = \Lambda$$

$$R_{11}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{2\,\mathcal{C}'}{a^2\,r\,\mathcal{C}} + \frac{(\mathcal{C}')^2}{2\,a^2\,\mathcal{C}^2} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,\mathcal{D}_5^2} - \frac{\mathcal{C}''}{a^2\,\mathcal{C}} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} - \frac{\mathcal{D}_5''}{2\,a^2\,\mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2\,r^2} + \frac{1}{a^2\,r^2\,\mathcal{C}} + \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} -$$
$$\frac{2\,\mathcal{C}'}{a^2\,r\,\mathcal{C}} - \frac{\mathcal{D}_4'}{2\,a^2\,r\,\mathcal{D}_4} - \frac{\mathcal{C}'\,\mathcal{D}_4'}{4\,a^2\,\mathcal{C}\,\mathcal{D}_4} - \frac{\mathcal{D}_5'}{2\,a^2\,r\,\mathcal{D}_5} - \frac{\mathcal{C}'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{C}\,\mathcal{D}_5} - \frac{\mathcal{C}''}{2\,a^2\,\mathcal{C}} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{\mathcal{D}_4'}{a^2\,r\,\mathcal{D}_4} - \frac{\mathcal{C}'\,\mathcal{D}_4'}{2\,a^2\,\mathcal{C}\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{D}_4\,\mathcal{D}_5} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{\mathcal{D}_5'}{a^2\,r\,\mathcal{D}_5} - \frac{\mathcal{C}'\,\mathcal{D}_5'}{2\,a^2\,\mathcal{C}\,\mathcal{D}_5} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,\mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\,a^2\,\mathcal{D}_5} = \Lambda$$

We solve R_{00} for $\mathcal{A}(t)$ and again get:

$$\mathcal{A}(t) = \frac{5\,c^2\,\dot{a}^2}{5\,c^2\,k + a^2\,\Lambda}$$ where k is a constant of integration.

Plugging this gives us 4 equations in the 3 unknowns $\mathcal{C}(r)$, $\mathcal{D}_4(r)$ and $\mathcal{D}_5(r)$:

$$R_{11}: \quad \frac{4\,c^2\,k}{a^2} + \Lambda - \frac{2\,\mathcal{C}'}{a^2\,r\,\mathcal{C}} + \frac{(\mathcal{C}')^2}{2\,a^2\,\mathcal{C}^2} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,\mathcal{D}_5^2} - \frac{\mathcal{C}''}{a^2\,\mathcal{C}} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} - \frac{\mathcal{D}_5''}{2\,a^2\,\mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{4\,c^2\,k}{a^2} - \frac{1}{a^2\,r^2} + \frac{1}{a^2\,r^2\,\mathcal{C}} + \Lambda - \frac{2\,\mathcal{C}'}{a^2\,r\,\mathcal{C}} - \frac{\mathcal{D}_4'}{2\,a^2\,r\,\mathcal{D}_4} - \frac{\mathcal{C}'\,\mathcal{D}_4'}{4\,a^2\,\mathcal{C}\,\mathcal{D}_4} - \frac{\mathcal{D}_5'}{2\,a^2\,r\,\mathcal{D}_5} - \frac{\mathcal{C}'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{C}\,\mathcal{D}_5} - \frac{\mathcal{C}''}{2\,a^2\,\mathcal{C}} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,k}{a^2} + \Lambda - \frac{\mathcal{D}_4'}{a^2\,r\,\mathcal{D}_4} - \frac{\mathcal{C}'\,\mathcal{D}_4'}{2\,a^2\,\mathcal{C}\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{D}_4\,\mathcal{D}_5} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{4\,c^2\,k}{a^2} + \Lambda - \frac{\mathcal{D}_5'}{a^2\,r\,\mathcal{D}_5} - \frac{\mathcal{C}'\,\mathcal{D}_5'}{2\,a^2\,\mathcal{C}\,\mathcal{D}_5} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,\mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\,a^2\,\mathcal{D}_5} = \Lambda$$

We can solve R_{44}-R_{55} for $\mathcal{C}$:

$$R_{44}\text{-}R_{55}: \quad -\frac{\mathcal{D}_4'}{a^2\,r\,\mathcal{D}_4} - \frac{\mathcal{C}'\,\mathcal{D}_4'}{2\,a^2\,\mathcal{C}\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} + \frac{\mathcal{D}_5'}{a^2\,r\,\mathcal{D}_5} + \frac{\mathcal{C}'\,\mathcal{D}_5'}{2\,a^2\,\mathcal{C}\,\mathcal{D}_5} - \frac{(\mathcal{D}_5')^2}{4\,a^2\,\mathcal{D}_5^2} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} + \frac{\mathcal{D}_5''}{2\,a^2\,\mathcal{D}_5} = 0$$

$$\mathcal{C}(r) = \frac{k\,\sqrt{\mathcal{D}_4}\,\sqrt{\mathcal{D}_5}}{r^2\,(\mathcal{D}_5\,\mathcal{D}_4' - \mathcal{D}_4\,\mathcal{D}_5')}$$

With this formula, R_{44} and R_{55} are dependent, so we are free to reduce the number of equations by one. However, the equations have become just too complex to work with.

aii: $\mathcal{B}(t,r)=a(t)^2\mathcal{B}(r)$, $\mathcal{C}(t,r)=a[t]^2$

Starting fresh, the resulting field equations are :

$$R_{00}: \quad \frac{5\,c^2\,\ddot{a}}{a\,A} - \frac{5\,c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{a^2\,A\,r\,B} + \frac{(A')^2}{4\,a^2\,A^2\,B} + \frac{A'\,B'}{4\,a^2\,A\,B^2} - \frac{A'\,D_4'}{4\,a^2\,A\,B\,D_4} - \frac{A'\,D_5'}{4\,a^2\,A\,B\,D_5} - \frac{A''}{2\,a^2\,A\,B} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}\,A'}{a\,A} = 0$$

$$R_{11}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{(A')^2}{4\,a^2\,A^2\,B} + \frac{B'}{a^2\,r\,B^2} + \frac{A'\,B'}{4\,a^2\,A\,B^2} +$$
$$\frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{A''}{2\,a^2\,A\,B} - \frac{D_4''}{2\,a^2\,B\,D_4} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{4\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{A'}{2\,a^2\,A\,r\,B} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{D_4'}{2\,a^2\,r\,B\,D_4} - \frac{D_5'}{2\,a^2\,r\,B\,D_5} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{D_4'}{a^2\,r\,B\,D_4} - \frac{A'\,D_4'}{4\,a^2\,A\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{D_5'}{a^2\,r\,B\,D_5} - \frac{A'\,D_5'}{4\,a^2\,A\,B\,D_5} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

From R_{01} we get that

$A(t,r) = \mathcal{A}[t]$ where $\mathcal{A}(t)$ is a new unknown.

R_{01} is eliminated, and the remaining equations become

$$R_{00}: \quad \frac{5\,c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{5\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} = \Lambda$$

R_{11}:

$$\frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{B'}{a^2\,r\,B^2} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_4''}{2\,a^2\,B\,D_4} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{1}{a^2\,r^2\,B} + \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} + \frac{B'}{2\,a^2\,r\,B^2} - \frac{D_4'}{2\,a^2\,r\,B\,D_4} - \frac{D_5'}{2\,a^2\,r\,B\,D_5} = \Lambda$$

$$R_{44}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{D_4'}{a^2\,r\,B\,D_4} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad \frac{4\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} - \frac{D_5'}{a^2\,r\,B\,D_5} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,a^2\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

We solve R_{00} for $\mathcal{A}(t)$ and again get:

$\mathcal{A}(t) = \dfrac{5\,c^2\,\dot{a}^2}{5\,c^2\,k + a^2\,\Lambda}$ where k is a constant of integration.

Plugging this in gives us 4 equations in the 3 unknowns $C(r)$, $D_4(r)$ and $D_5(r)$:

$$R_{11}: \quad \frac{4\,c^2\,k}{a^2} + \Lambda + \frac{B'}{a^2\,r\,B^2} + \frac{B'\,D_4'}{4\,a^2\,B^2\,D_4} + \frac{(D_4')^2}{4\,a^2\,B\,D_4^2} + \frac{B'\,D_5'}{4\,a^2\,B^2\,D_5} + \frac{(D_5')^2}{4\,a^2\,B\,D_5^2} - \frac{D_4''}{2\,a^2\,B\,D_4} - \frac{D_5''}{2\,a^2\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{4 c^2 k}{a^2} + \frac{1}{a^2 r^2} - \frac{1}{a^2 r^2 B} + \Lambda + \frac{B'}{2 a^2 r B^2} - \frac{D_4'}{2 a^2 r B D_4} - \frac{D_5'}{2 a^2 r B D_5} = \Lambda$$

$$R_{44}: \quad \frac{4 c^2 k}{a^2} + \Lambda - \frac{D_4'}{a^2 r B D_4} + \frac{B' D_4'}{4 a^2 B^2 D_4} + \frac{(D_4')^2}{4 a^2 B D_4^2} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} - \frac{D_4''}{2 a^2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{4 c^2 k}{a^2} + \Lambda - \frac{D_5'}{a^2 r B D_5} + \frac{B' D_5'}{4 a^2 B^2 D_5} - \frac{D_4' D_5'}{4 a^2 B D_4 D_5} + \frac{(D_5')^2}{4 a^2 B D_5^2} - \frac{D_5''}{2 a^2 B D_5} = \Lambda$$

At this point, there are many different approaches that can be tried. I pursued many of them and I show a few of them below, but none of them led to a complete solution so the reader may wish to skip to the next case.

We can solve $R_{11}+R_{44}-R_{55}$ for any of the three unknowns. Solving for D_5 turns out to be the most productive:

$R_{11}+R_{44}-R_{55}$:

$$-\frac{D_4'}{a^2 r D_4} - \frac{C' D_4'}{2 a^2 C D_4} + \frac{(D_4')^2}{4 a^2 D_4^2} + \frac{D_5'}{a^2 r D_5} + \frac{C' D_5'}{2 a^2 C D_5} - \frac{(D_5')^2}{4 a^2 D_5^2} - \frac{D_4''}{2 a^2 D_4} + \frac{D_5''}{2 a^2 D_5} = \text{rhs45}$$

$$D_5(r) = e^{\int_1^r \frac{-8 c^2 k B^2 \rho D_4^2 - 2 D_4^2 B' + 2 B D_4 D_4' - \rho D_4 B' D_4' - B \rho (D_4')^2 + 2 B \rho D_4 D_4''}{2 B D_4^2} d\rho} \zeta$$

where ζ is a constant of integration.

Note that $g_{55}=-a(t)^2 D_4(r) p_5^2$. ζ can be negative or positive, so $\zeta p_5^2 \equiv \pm \hat{p}_5^2$, and $\hat{p}_5$ is just another radius parameter. So, we can continue solving the equations using the new radius $\hat{p}_5$ or, better yet, rename $\hat{p}_5$ back to p_5. I describe this process as 'rolling ζ into the radius p_5':

$$D_5(r) = \pm e^{\int_1^r \frac{-8 c^2 k B^2 \rho D_4^2 - 2 D_4^2 B' + 2 B D_4 D_4' - \rho D_4 B' D_4' - B \rho (D_4')^2 + 2 B \rho D_4 D_4''}{2 B D_4^2} d\rho}$$

Plugging this back in, the integral conveniently disappears from all of the equations. R_{11} becomes a bit lengthy, but the other two equations become:

$$R_{22}: \quad \frac{6 c^2 k}{a^2} + \frac{1}{a^2 r^2} - \frac{1}{a^2 r^2 B} + \Lambda + \frac{B'}{a^2 r B^2} - \frac{D_4'}{a^2 r B D_4} + \frac{B' D_4'}{4 a^2 B^2 D_4} + \frac{(D_4')^2}{4 a^2 B D_4^2} - \frac{D_4''}{2 a^2 B D_4} = \Lambda$$

$$R_{44}: \quad \frac{4 c^2 k}{a^2} + \Lambda + \frac{c^2 k r D_4'}{a^2 D_4} - \frac{D_4'}{a^2 r B D_4} + \frac{B' D_4'}{2 a^2 B^2 D_4} + \frac{r B' (D_4')^2}{8 a^2 B^2 D_4^2} + \frac{r (D_4')^3}{8 a^2 B D_4^3} - \frac{D_4''}{2 a^2 B D_4} - \frac{r D_4' D_4''}{4 a^2 B D_4^2} = \Lambda$$

R_{22} can be solved for B (but not for D_4), but the formula involves an integral and a double integral, and even though the integrals (again) conveniently disappear in R_{44}, it is too difficult to solve that equation for D_4. Moreover, R_{11} becomes even more complex.

Instead of solving $R_{11}+R_{44}-R_{55}$ for D_5, we can solve $R_{11}-2R_{22}-R_{44}-R_{55}$ for D_5:

$$R_{11}-2R_{22}-R_{44}-R_{55}:\quad -\frac{12\,c^2\,k}{a^2}-\frac{2}{a^2\,r^2}+\frac{2}{a^2\,r^2\,B}-3\,\Lambda+\frac{2\,D_4'}{a^2\,r\,B\,D_4}+\frac{2\,D_5'}{a^2\,r\,B\,D_5}+\frac{D_4'\,D_5'}{2\,a^2\,B\,D_4\,D_5}=-3\,\Lambda$$

$$D_5(r)=e^{\int_1^r \frac{4\left(-D_4+B\,D_4+6\,c^2\,k\,B\,\rho^2\,D_4-\rho\,D_4'\right)}{\rho\,(4\,D_4+\rho\,D_4')}\,d\rho}\,\alpha$$ where α is a constant of integration.

With this formula, these 4 equations are dependent and we are free to designate R_{11} as redundant. The other equations simplify nicely:

$$R_{22}:\quad \frac{4\,c^2\,k\,D_4}{a^2\,(4\,D_4+r\,D_4')}+\frac{2\,D_4}{a^2\,r^2\,(4\,D_4+r\,D_4')}-\frac{2\,D_4}{a^2\,r^2\,B\,(4\,D_4+r\,D_4')}+\frac{4\,\Lambda\,D_4}{4\,D_4+r\,D_4'}+\frac{2\,D_4\,B'}{a^2\,r\,B^2\,(4\,D_4+r\,D_4')}+\frac{D_4'}{a^2\,r\,(4\,D_4+r\,D_4')}+\frac{4\,c^2\,k\,r\,D_4'}{a^2\,(4\,D_4+r\,D_4')}-\frac{D_4'}{a^2\,r\,B\,(4\,D_4+r\,D_4')}+\frac{r\,\Lambda\,D_4'}{4\,D_4+r\,D_4'}+\frac{B'\,D_4'}{2\,a^2\,B^2\,(4\,D_4+r\,D_4')}-\frac{(D_4')^2}{2\,a^2\,B\,D_4\,(4\,D_4+r\,D_4')}=\Lambda$$

$$R_{44}:\quad \frac{16\,c^2\,k\,D_4}{a^2\,(4\,D_4+r\,D_4')}+\frac{4\,\Lambda\,D_4}{4\,D_4+r\,D_4'}-\frac{D_4'}{a^2\,r\,(4\,D_4+r\,D_4')}-\frac{2\,c^2\,k\,r\,D_4'}{a^2\,(4\,D_4+r\,D_4')}-\frac{3\,D_4'}{a^2\,r\,B\,(4\,D_4+r\,D_4')}+\frac{r\,\Lambda\,D_4'}{4\,D_4+r\,D_4'}+\frac{B'\,D_4'}{a^2\,B^2\,(4\,D_4+r\,D_4')}+\frac{(D_4')^2}{a^2\,B\,D_4\,(4\,D_4+r\,D_4')}+\frac{r\,B'\,(D_4')^2}{4\,a^2\,B^2\,D_4\,(4\,D_4+r\,D_4')}+\frac{r\,(D_4')^3}{4\,a^2\,B\,D_4^2\,(4\,D_4+r\,D_4')}-\frac{2\,D_4''}{a^2\,B\,(4\,D_4+r\,D_4')}-\frac{r\,D_4'\,D_4''}{2\,a^2\,B\,D_4\,(4\,D_4+r\,D_4')}=\Lambda$$

$$R_{55}:\quad -\frac{32\,c^2\,k\,D_4^2}{a^2\,(4\,D_4+r\,D_4')^2}+\frac{4\,D_4^2}{a^2\,r^2\,B\,(4\,D_4+r\,D_4')^2}-\frac{48\,c^2\,k\,B\,D_4^2}{a^2\,(4\,D_4+r\,D_4')^2}-\frac{4\,B\,D_4^2}{a^2\,r^2\,(4\,D_4+r\,D_4')^2}-\frac{144\,c^4\,k^2\,r^2\,B\,D_4^2}{a^2\,(4\,D_4+r\,D_4')^2}+\frac{16\,\Lambda\,D_4^2}{(4\,D_4+r\,D_4')^2}-\frac{4\,D_4^2\,B'}{a^2\,r\,B^2\,(4\,D_4+r\,D_4')^2}-\frac{4\,D_4^2\,B'}{a^2\,r\,B\,(4\,D_4+r\,D_4')^2}-\frac{24\,c^2\,k\,r\,D_4^2\,B'}{a^2\,B\,(4\,D_4+r\,D_4')^2}+\frac{4\,D_4\,D_4'}{a^2\,r\,(4\,D_4+r\,D_4')^2}+\frac{32\,c^2\,k\,r\,D_4\,D_4'}{a^2\,(4\,D_4+r\,D_4')^2}+\frac{12\,D_4\,D_4'}{a^2\,r\,B\,(4\,D_4+r\,D_4')^2}+\frac{8\,r\,\Lambda\,D_4\,D_4'}{(4\,D_4+r\,D_4')^2}-\frac{5\,D_4\,B'\,D_4'}{a^2\,B^2\,(4\,D_4+r\,D_4')^2}-\frac{D_4\,B'\,D_4'}{a^2\,B\,(4\,D_4+r\,D_4')^2}-\frac{6\,c^2\,k\,r^2\,D_4\,B'\,D_4'}{a^2\,B\,(4\,D_4+r\,D_4')^2}-\frac{3\,(D_4')^2}{a^2\,(4\,D_4+r\,D_4')^2}-\frac{14\,c^2\,k\,r^2\,(D_4')^2}{a^2\,(4\,D_4+r\,D_4')^2}-\frac{3\,(D_4')^2}{a^2\,B\,(4\,D_4+r\,D_4')^2}+\frac{r^2\,\Lambda\,(D_4')^2}{(4\,D_4+r\,D_4')^2}-\frac{r\,B'\,(D_4')^2}{a^2\,B^2\,(4\,D_4+r\,D_4')^2}+\frac{r\,(D_4')^3}{a^2\,B\,D_4\,(4\,D_4+r\,D_4')^2}+\frac{2\,D_4\,D_4''}{a^2\,(4\,D_4+r\,D_4')^2}+\frac{12\,c^2\,k\,r^2\,D_4\,D_4''}{a^2\,(4\,D_4+r\,D_4')^2}+\frac{6\,D_4\,D_4''}{a^2\,B\,(4\,D_4+r\,D_4')^2}=\Lambda$$

The solution for D_4 using R_{22} is an even more complex antiderivative, and while the antiderivative conveniently disappears when plugging it back in to R_{44} and R_{55}, those equations are extremely long and complex. There is no hope of solving them for the remaining unknown, B.

Another try: instead of solving $R_{11}-2R_{22}-R_{44}-R_{55}$ for D_5, this equation can be algebraically solved for B, but the solution is complex and makes the resulting equations even more complex when it is substituted back in.

$$B(r) = \frac{4\,D_4\,D_5 + 4\,r\,D_5\,D_4' + 4\,r\,D_4\,D_5' + r^2\,D_4'\,D_5'}{4\left(1 + 6\,c^2\,k\,r^2\right)D_4\,D_5}$$

I also tried other combinations of the field equations, all to no avail.

The field equations individually are also difficult to solve. The analogous equations were solvable for D_4 in the 5D case, but the existence of another curved-space metric, D_5, makes these equations much more complex.

These equations are too complex for me to solve.

b: B(t,r)=B(r), C(t,r)=C(r)

The field equations become:

$$R_{00}:\quad \frac{2\,c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{a\,A^2} - \frac{A'}{A\,r\,B} + \frac{(A')^2}{4\,A^2\,B} - \frac{A'\,C'}{2\,A\,C\,B} + \frac{A'\,B'}{4\,A\,B^2} - \frac{A'\,D_4'}{4\,A\,B\,D_4} - \frac{A'\,D_5'}{4\,A\,B\,D_5} - \frac{A''}{2\,A\,B} = \Lambda$$

$$R_{01}:\quad -\frac{\dot{a}\,A'}{a\,A} + \frac{\dot{a}\,D_4'}{2\,a\,D_4} + \frac{\dot{a}\,D_5'}{2\,a\,D_5} = 0$$

$$R_{11}:\quad \frac{(A')^2}{4\,A^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} + \frac{B'}{r\,B^2} + \frac{A'\,B'}{4\,A\,B^2} + \frac{C'\,B'}{2\,C\,B^2} +$$
$$\frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{4\,B\,D_4^2} + \frac{B'\,D_5'}{4\,B^2\,D_5} + \frac{(D_5')^2}{4\,B\,D_5^2} - \frac{A''}{2\,A\,B} - \frac{C''}{C\,B} - \frac{D_4''}{2\,B\,D_4} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

$$R_{22}:\quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{A'}{2\,A\,r\,B} - \frac{2\,C'}{r\,C\,B} - \frac{A'\,C'}{4\,A\,C\,B} +$$
$$\frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{D_4'}{2\,r\,B\,D_4} - \frac{C'\,D_4'}{4\,C\,B\,D_4} - \frac{D_5'}{2\,r\,B\,D_5} - \frac{C'\,D_5'}{4\,C\,B\,D_5} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}:\quad \frac{c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{D_4'}{r\,B\,D_4} - \frac{A'\,D_4'}{4\,A\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{4\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{55}:\quad \frac{c^2\,\dot{a}^2}{a^2\,A} + \frac{c^2\,\ddot{a}}{a\,A} - \frac{c^2\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{D_5'}{r\,B\,D_5} - \frac{A'\,D_5'}{4\,A\,B\,D_5} - \frac{C'\,D_5'}{2\,C\,B\,D_5} + \frac{B'\,D_5'}{4\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,B\,D_5^2} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

Because $a(t) \neq$ constant, $\dot{a} \neq 0$. This allows us to simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then we can solve for A:

$$R_{01}:\quad \frac{A'}{A} - \frac{D_4'}{2\,D_4} - \frac{D_5'}{2\,D_5} = 0$$

$A(t,r) = \mathcal{A}[t]\,\sqrt{D_4[r]}\,\sqrt{D_5[r]}$ where $\mathcal{A}(t)$ is a new unknown.

The equations now become

$$R_{00}: \quad \frac{2 c^2 \ddot{a}}{a \mathcal{A} \sqrt{D_4} \sqrt{D_5}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2 \sqrt{D_4} \sqrt{D_5}} - \frac{D_4'}{2 r B D_4} - \frac{C' D_4'}{4 C B D_4} + \frac{B' D_4'}{8 B^2 D_4} + \frac{(D_4')^2}{16 B D_4^2} - \frac{D_5'}{2 r B D_5} - \frac{C' D_5'}{4 C B D_5} + \frac{B' D_5'}{8 B^2 D_5} - \frac{3 D_4' D_5'}{8 B D_4 D_5} + \frac{(D_5')^2}{16 B D_5^2} - \frac{D_4''}{4 B D_4} - \frac{D_5''}{4 B D_5} = \Lambda$$

$$R_{11}: \quad -\frac{2 C'}{r C B} + \frac{(C')^2}{2 C^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2 C B^2} + \frac{3 B' D_4'}{8 B^2 D_4} + \frac{7 (D_4')^2}{16 B D_4^2} + \frac{3 B' D_5'}{8 B^2 D_5} - \frac{D_4' D_5'}{8 B D_4 D_5} + \frac{7 (D_5')^2}{16 B D_5^2} - \frac{C''}{C B} - \frac{3 D_4''}{4 B D_4} - \frac{3 D_5''}{4 B D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2 C'}{r C B} + \frac{B'}{2 r B^2} + \frac{C' B'}{4 C B^2} - \frac{3 D_4'}{4 r B D_4} - \frac{3 C' D_4'}{8 C B D_4} - \frac{3 D_5'}{4 r B D_5} - \frac{3 C' D_5'}{8 C B D_5} - \frac{C''}{2 C B} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \dot{a}^2}{a^2 \mathcal{A} \sqrt{D_4} \sqrt{D_5}} + \frac{c^2 \ddot{a}}{a \mathcal{A} \sqrt{D_4} \sqrt{D_5}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \sqrt{D_4} \sqrt{D_5}} - \frac{D_4'}{r B D_4} - \frac{C' D_4'}{2 C B D_4} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{8 B D_4^2} - \frac{3 D_4' D_5'}{8 B D_4 D_5} - \frac{D_4''}{2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2 \dot{a}^2}{a^2 \mathcal{A} \sqrt{D_4} \sqrt{D_5}} + \frac{c^2 \ddot{a}}{a \mathcal{A} \sqrt{D_4} \sqrt{D_5}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \sqrt{D_4} \sqrt{D_5}} - \frac{D_5'}{r B D_5} - \frac{C' D_5'}{2 C B D_5} + \frac{B' D_5'}{4 B^2 D_5} - \frac{3 D_4' D_5'}{8 B D_4 D_5} + \frac{(D_5')^2}{8 B D_5^2} - \frac{D_5''}{2 B D_5} = \Lambda$$

From $2R_{00}-R_{44}-R_{55}$ we can solve for $\mathcal{A}(t)$:

$$2R_{00}-R_{44}-R_{55}: \quad -\frac{2 c^2 \dot{a}^2}{a^2 \mathcal{A} \sqrt{D_4} \sqrt{D_5}} + \frac{2 c^2 \ddot{a}}{a \mathcal{A} \sqrt{D_4} \sqrt{D_5}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2 \sqrt{D_4} \sqrt{D_5}} = 0$$

$$\Rightarrow \quad 2R_{00}-R_{44}-R_{55}: \quad -2 c^2 \mathcal{A} \dot{a}^2 + 2 a c^2 \mathcal{A} \ddot{a} - a c^2 \dot{a} \dot{\mathcal{A}} = 0$$

$$\mathcal{A}(t) = \frac{k \dot{a}[t]^2}{a[t]^2}$$ where k is a constant of integration.

With this formula, R_{00}, R_{44}, and R_{55} are dependent equations, so we can choose to designate R_{00} as redundant.

The remaining equations become:

$$R_{11}: \quad -\frac{2 C'}{r C B} + \frac{(C')^2}{2 C^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2 C B^2} + \frac{3 B' D_4'}{8 B^2 D_4} + \frac{7 (D_4')^2}{16 B D_4^2} + \frac{3 B' D_5'}{8 B^2 D_5} - \frac{D_4' D_5'}{8 B D_4 D_5} + \frac{7 (D_5')^2}{16 B D_5^2} - \frac{C''}{C B} - \frac{3 D_4''}{4 B D_4} - \frac{3 D_5''}{4 B D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2 C'}{r C B} + \frac{B'}{2 r B^2} + \frac{C' B'}{4 C B^2} - \frac{3 D_4'}{4 r B D_4} - \frac{3 C' D_4'}{8 C B D_4} - \frac{3 D_5'}{4 r B D_5} - \frac{3 C' D_5'}{8 C B D_5} - \frac{C''}{2 C B} = \Lambda$$

$$R_{44}: \quad \frac{2 c^2}{k \sqrt{D_4} \sqrt{D_5}} - \frac{D_4'}{r B D_4} - \frac{C' D_4'}{2 C B D_4} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{8 B D_4^2} - \frac{3 D_4' D_5'}{8 B D_4 D_5} - \frac{D_4''}{2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{2 c^2}{k \sqrt{D_4} \sqrt{D_5}} - \frac{D_5'}{r B D_5} - \frac{C' D_5'}{2 C B D_5} + \frac{B' D_5'}{4 B^2 D_5} - \frac{3 D_4' D_5'}{8 B D_4 D_5} + \frac{(D_5')^2}{8 B D_5^2} - \frac{D_5''}{2 B D_5} = \Lambda$$

I tried to solve several of the equations, as well as several algebraic combinations of the equations, for each of the unknowns, but could not find a solution that would satisfy more than just a single equation.

We next examine various subcases of Case b that add assumptions upfront that could change the outcome of the field equations.

b_Iβ: A(t,r)=A(r)

b_II: A(t,r)=c²

bii: 𝕔(t,r)=1

biii: B(t,r)=1 and 𝕔(t,r)=1

b_Iβ: B(t,r)=B(r), 𝕔(t,r)=C(r), A(t,r)=A(r)

We try Case b again with the additional assumption that A(t,r)=A(r). Starting fresh, the field equations are identical to those in Case b except that all instances of A(t,r) are replaced by A(r):

$$R_{00}: \quad \frac{2 c^2 \ddot{a}}{a A} - \frac{A'}{r A B} - \frac{C' A'}{2 C A B} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' D_4'}{4 A B D_4} - \frac{A' D_5'}{4 A B D_5} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a} A'}{a A} + \frac{\dot{a} D_4'}{2 a D_4} + \frac{\dot{a} D_5'}{2 a D_5} = 0$$

$$R_{11}: \quad -\frac{2 C'}{r C B} + \frac{(C')^2}{2 C^2 B} + \frac{(A')^2}{4 A^2 B} + \frac{B'}{r B^2} + \frac{C' B'}{2 C B^2} + \frac{A' B'}{4 A B^2} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{4 B D_4^2} + \frac{B' D_5'}{4 B^2 D_5} + \frac{(D_5')^2}{4 B D_5^2} - \frac{C''}{C B} - \frac{A''}{2 A B} - \frac{D_4''}{2 B D_4} - \frac{D_5''}{2 B D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2 C} - \frac{1}{r^2 B} - \frac{2 C'}{r C B} - \frac{A'}{2 r A B} - \frac{C' A'}{4 C A B} + \frac{B'}{2 r B^2} + \frac{C' B'}{4 C B^2} - \frac{D_4'}{2 r B D_4} - \frac{C' D_4'}{4 C B D_4} - \frac{D_5'}{2 r B D_5} - \frac{C' D_5'}{4 C B D_5} - \frac{C''}{2 C B} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D_4'}{r B D_4} - \frac{C' D_4'}{2 C B D_4} - \frac{A' D_4'}{4 A B D_4} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{4 B D_4^2} - \frac{D_4' D_5'}{4 B D_4 D_5} - \frac{D_4''}{2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D_5'}{r B D_5} - \frac{C' D_5'}{2 C B D_5} - \frac{A' D_5'}{4 A B D_5} + \frac{B' D_5'}{4 B^2 D_5} - \frac{D_4' D_5'}{4 B D_4 D_5} + \frac{(D_5')^2}{4 B D_5^2} - \frac{D_5''}{2 B D_5} = \Lambda$$

Like Case b, we can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \quad \frac{A'}{A} - \frac{D_4'}{2 D_4} - \frac{D_5'}{2 D_5} = 0$$

$A(r) = \zeta \sqrt{D_4} \sqrt{D_5}$ where ζ is a constant of integration.

The equations now become

$$R_{00}: \quad \frac{2\,c^2\,\ddot{a}}{a\,\zeta\,\sqrt{D_4}\,\sqrt{D_5}} - \frac{D_4'}{2\,r\,B\,D_4} - \frac{C'\,D_4'}{4\,C\,B\,D_4} + \frac{B'\,D_4'}{8\,B^2\,D_4} + \frac{(D_4')^2}{16\,B\,D_4^2} - \frac{D_5'}{2\,r\,B\,D_5} - \frac{C'\,D_5'}{4\,C\,B\,D_5} + \frac{B'\,D_5'}{8\,B^2\,D_5} - \frac{3\,D_4'\,D_5'}{8\,B\,D_4\,D_5} + \frac{(D_5')^2}{16\,B\,D_5^2} - \frac{D_4''}{4\,B\,D_4} - \frac{D_5''}{4\,B\,D_5} = \Lambda$$

$$R_{11}: \quad -\frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} + \frac{B'}{r\,B^2} + \frac{C'\,B'}{2\,C\,B^2} + \frac{3\,B'\,D_4'}{8\,B^2\,D_4} + \frac{7\,(D_4')^2}{16\,B\,D_4^2} + \frac{3\,B'\,D_5'}{8\,B^2\,D_5} - \frac{D_4'\,D_5'}{8\,B\,D_4\,D_5} + \frac{7\,(D_5')^2}{16\,B\,D_5^2} - \frac{C''}{C\,B} - \frac{3\,D_4''}{4\,B\,D_4} - \frac{3\,D_5''}{4\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{3\,D_4'}{4\,r\,B\,D_4} - \frac{3\,C'\,D_4'}{8\,C\,B\,D_4} - \frac{3\,D_5'}{4\,r\,B\,D_5} - \frac{3\,C'\,D_5'}{8\,C\,B\,D_5} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,\zeta\,\sqrt{D_4}\,\sqrt{D_5}} + \frac{c^2\,\ddot{a}}{a\,\zeta\,\sqrt{D_4}\,\sqrt{D_5}} - \frac{D_4'}{r\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{8\,B\,D_4^2} - \frac{3\,D_4'\,D_5'}{8\,B\,D_4\,D_5} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2\,\dot{a}^2}{a^2\,\zeta\,\sqrt{D_4}\,\sqrt{D_5}} + \frac{c^2\,\ddot{a}}{a\,\zeta\,\sqrt{D_4}\,\sqrt{D_5}} - \frac{D_5'}{r\,B\,D_5} - \frac{C'\,D_5'}{2\,C\,B\,D_5} + \frac{B'\,D_5'}{4\,B^2\,D_5} - \frac{3\,D_4'\,D_5'}{8\,B\,D_4\,D_5} + \frac{(D_5')^2}{8\,B\,D_5^2} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

$2R_{00}-R_{44}-R_{55}$ can be solved for $a(t)$:

$$2R_{00}-R_{44}-R_{55}: \quad -\frac{2\,c^2\,\dot{a}^2}{a^2\,\zeta\,\sqrt{D_4}\,\sqrt{D_5}} + \frac{2\,c^2\,\ddot{a}}{a\,\zeta\,\sqrt{D_4}\,\sqrt{D_5}} = 0$$

$a(t) = e^{k\,t}\,\alpha$ where α and k are constants of integration.

Plugging this in, we can declare R_{00} redundant,

equal to $\frac{R_{44}+R_{55}}{2}$, and the equations become slightly simplified:

$$R_{11}: \quad -\frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} + \frac{B'}{r\,B^2} + \frac{C'\,B'}{2\,C\,B^2} + \frac{3\,B'\,D_4'}{8\,B^2\,D_4} + \frac{7\,(D_4')^2}{16\,B\,D_4^2} + \frac{3\,B'\,D_5'}{8\,B^2\,D_5} - \frac{D_4'\,D_5'}{8\,B\,D_4\,D_5} + \frac{7\,(D_5')^2}{16\,B\,D_5^2} - \frac{C''}{C\,B} - \frac{3\,D_4''}{4\,B\,D_4} - \frac{3\,D_5''}{4\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{3\,D_4'}{4\,r\,B\,D_4} - \frac{3\,C'\,D_4'}{8\,C\,B\,D_4} - \frac{3\,D_5'}{4\,r\,B\,D_5} - \frac{3\,C'\,D_5'}{8\,C\,B\,D_5} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}: \quad \frac{2\,c^2\,k^2}{\zeta\,\sqrt{D_4}\,\sqrt{D_5}} - \frac{D_4'}{r\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{8\,B\,D_4^2} - \frac{3\,D_4'\,D_5'}{8\,B\,D_4\,D_5} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad \frac{2\,c^2\,k^2}{\zeta\,\sqrt{D_4}\,\sqrt{D_5}} - \frac{D_5'}{r\,B\,D_5} - \frac{C'\,D_5'}{2\,C\,B\,D_5} + \frac{B'\,D_5'}{4\,B^2\,D_5} - \frac{3\,D_4'\,D_5'}{8\,B\,D_4\,D_5} + \frac{(D_5')^2}{8\,B\,D_5^2} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

$R_{44}-R_{55}$ eliminates the D_4D_5 product term and can be solved for C:

$$R_{44}-R_{55}: \quad -\frac{D_4'}{r\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{8\,B\,D_4^2} + \frac{D_5'}{r\,B\,D_5} + \frac{C'\,D_5'}{2\,C\,B\,D_5} - \frac{B'\,D_5'}{4\,B^2\,D_5} - \frac{(D_5')^2}{8\,B\,D_5^2} - \frac{D_4''}{2\,B\,D_4} + \frac{D_5''}{2\,B\,D_5} = 0$$

$C(r) = \frac{\beta\,\sqrt{B}\,D_4^{1/4}\,D_5^{1/4}}{r^2\,(D_5\,D_4' - D_4\,D_5')}$ where β is a constant of integration.

Plugging $C(r)$ back in makes R_{11} and R_{22} too lengthy to display here, and equations R_{44} and R_{55} are identical, redundant:

$$R_{44}: \quad \frac{2 c^2 k^2}{\zeta \sqrt{D_4} \sqrt{D_5}} - \frac{D_4' D_5'}{2 B D_4 D_5} + \frac{D_5' D_4''}{2 (B D_5 D_4' - B D_4 D_5')} - \frac{D_4' D_5''}{2 (B D_5 D_4' - B D_4 D_5')} = \Lambda$$

R_{44} can be algebraically solved for B:

$$B(r) = \frac{-\zeta D_5 (D_4')^2 D_5' + \zeta D_4 D_4' (D_5')^2 + \zeta D_4 D_5 D_5' D_4'' - \zeta D_4 D_5 D_4' D_5''}{2 \left(-2 c^2 k^2 \sqrt{D_4} D_5^{3/2} D_4' + \zeta \Lambda D_4 D_5^2 D_4' + 2 c^2 k^2 D_4^{3/2} \sqrt{D_5} D_5' - \zeta \Lambda D_4^2 D_5 D_5'\right)}.$$

But, the two remaining equations are pages and pages long. We cannot solve them.

Abandoning $B(r)$ and $C(r)$, reverting back to the four equations R_{11}, R_{22}, R_{44}, and R_{55}, we find that $2R_{11}-4R_{22}-3R_{44}-3R_{55}$ is a combination of the equations that eliminates all second derivatives. It can be algebraically solved for B, but the the result is unwieldy and makes the remaining equations even more so.

$$2R_{11}-4R_{22}-3R_{44}-3R_{55}: \quad -\frac{4}{r^2 C} + \frac{4}{r^2 B} - \frac{12 c^2 k^2}{\zeta \sqrt{D_4} \sqrt{D_5}} + \frac{4 C'}{r C B} + \frac{(C')^2}{C^2 B} + \frac{6 D_4'}{r B D_4} + \frac{3 C' D_4'}{C B D_4} + \frac{(D_4')^2}{2 B D_4^2} + \frac{6 D_5'}{r B D_5} + \frac{3 C' D_5'}{C B D_5} + \frac{2 D_4' D_5'}{B D_4 D_5} + \frac{(D_5')^2}{2 B D_5^2} = -8 \Lambda$$

Instead of this, we observe that R_{44} retains a second derivative but can nonetheless be solved for C:

$$C(r) = e^{\int_1^r \frac{16 c^2 k^2 B^2 \rho D_4^{3/2} \sqrt{D_5} - 8 B^2 \zeta \Lambda \rho D_4^2 D_5 - 8 B \zeta D_4 D_5 D_4' + 2 \zeta \rho D_4 D_5 B' D_4' + B \zeta \rho D_5 (D_4')^2 - 3 B \zeta \rho D_4 D_4' D_5' - 4 B \zeta \rho D_4 D_5 D_4''}{4 B \zeta \rho D_4 D_5 D_4'} d\rho} \gamma$$

where γ is a constant of integration.

This makes R_{44} an identity, and the remaining equations are less complex than expected, most not involving the integral:

$$R_{11}: \quad \frac{5 \Lambda}{2} - \frac{3 c^2 k^2}{\zeta \sqrt{D_4} \sqrt{D_5}} + \frac{5 (B')^2}{8 B^3} - \frac{2 B \Lambda^2 D_4^2}{(D_4')^2} - \frac{8 c^4 k^4 B D_4}{\zeta^2 D_5 (D_4')^2} + \frac{8 c^2 k^2 B \Lambda D_4^{3/2}}{\zeta \sqrt{D_5} (D_4')^2} + \frac{D_4'''}{B D_4'} + \frac{2 \Lambda D_4 B'}{B D_4'} - \frac{4 c^2 k^2 \sqrt{D_4} B'}{B \zeta \sqrt{D_5} D_4'} + \frac{3 B' D_4'}{8 B^2 D_4} + \frac{21 (D_4')^2}{32 B D_4^2} + \frac{3 B' D_5'}{8 B^2 D_5} + \frac{5 c^2 k^2 \sqrt{D_4} D_5'}{\zeta D_5^{3/2} D_4'} - \frac{3 \Lambda D_4 D_5'}{2 D_5 D_4'} + \frac{D_4' D_5'}{16 B D_4 D_5} - \frac{19 (D_5')^2}{32 B D_5^2} - \frac{B''}{2 B^2} - \frac{3 D_4''}{4 B D_4} - \frac{4 \Lambda D_4 D_4''}{(D_4')^2} + \frac{8 c^2 k^2 \sqrt{D_4} D_4''}{\zeta \sqrt{D_5} (D_4')^2} - \frac{3 D_5' D_4''}{4 B D_5 D_4'} - \frac{3 (D_4'')^2}{2 B (D_4')^2} = \Lambda$$

$$R_{22}: \quad \frac{e^{-\int_1^r \frac{1}{4}\left(-\frac{8}{\rho}+\frac{2B'}{B}-\frac{B\left(8\zeta\Lambda D_4-\frac{16c^2k^2\sqrt{D_4}}{\sqrt{D_5}}\right)}{\zeta D_4'}+\frac{D_4'}{D_4}-\frac{3D_5'}{D_5}-\frac{4D_4''}{D_4'}\right)d\rho}}{r^2\gamma}+\frac{9\Lambda}{4}-\frac{7c^2k^2}{2\zeta\sqrt{D_4}\sqrt{D_5}}+\frac{(B')^2}{4B^3}-\frac{2B\Lambda^2D_4^2}{(D_4')^2}-\frac{8c^4k^4BD_4}{\zeta^2D_5(D_4')^2}+$$

$$\frac{8c^2k^2B\Lambda D_4^{3/2}}{\zeta\sqrt{D_5}(D_4')^2}+\frac{D_4'''}{2BD_4'}+\frac{3\Lambda D_4B'}{2BD_4'}-\frac{3c^2k^2\sqrt{D_4}B'}{B\zeta\sqrt{D_5}D_4'}-\frac{B'D_4'}{4B^2D_4}+\frac{5c^2k^2\sqrt{D_4}D_5'}{2\zeta D_5^{3/2}D_4'}-\frac{3\Lambda D_4D_5'}{4D_5D_4'}+\frac{3D_4'D_5'}{8BD_4D_5}-$$

$$\frac{3(D_5')^2}{8BD_5^2}-\frac{B''}{4B^2}+\frac{D_4''}{2BD_4}-\frac{3\Lambda D_4D_4''}{(D_4')^2}+\frac{6c^2k^2\sqrt{D_4}D_4''}{\zeta\sqrt{D_5}(D_4')^2}+\frac{B'D_4''}{4B^2D_4'}-\frac{3D_5'D_4''}{8BD_5D_4'}-\frac{(D_4'')^2}{B(D_4')^2}+\frac{3D_5''}{8BD_5}=\Lambda$$

$$R_{55}: \quad \frac{2c^2k^2}{\zeta\sqrt{D_4}\sqrt{D_5}}-\frac{2c^2k^2\sqrt{D_4}D_5'}{\zeta D_5^{3/2}D_4'}+\frac{\Lambda D_4D_5'}{D_5D_4'}-\frac{D_4'D_5'}{2BD_4D_5}+\frac{(D_5')^2}{2BD_5^2}+\frac{D_5'D_4''}{2BD_5D_4'}-\frac{D_5''}{2BD_5}=\Lambda$$

R_{55} can be algebraically solved for B:

$$B(r) = \frac{\zeta\left(-D_4D_4'(D_5')^2+D_5\left((D_4')^2D_5'-D_4D_5'D_4''+D_4D_4'D_5''\right)\right)}{2\sqrt{D_4}\left(-2c^2k^2+\zeta\Lambda\sqrt{D_4}\sqrt{D_5}\right)\sqrt{D_5}\left(-D_5D_4'+D_4D_5'\right)}$$

Unfortunately, this solution makes R_{11} much more unmanageable. Had we found a solution for R_{11}, then there would have been a chance to solve R_{22}, integral and all, because the equation would have only a single unknown.

This case is also too difficult for me to solve.

b_II: B(t,r)=B(r), C(t,r)=C(r), A(t,r)=c²

We try Case b yet again with the stronger assumption that $A(t,r)=c^2$. Starting fresh, the field equations are:

$$R_{00}: \quad \frac{2\ddot{a}}{a}=\Lambda$$

$$R_{01}: \quad \frac{\dot{a}D_4'}{2aD_4}+\frac{\dot{a}D_5'}{2aD_5}=0$$

$$R_{11}: \quad -\frac{2C'}{rCB}+\frac{(C')^2}{2C^2B}+\frac{B'}{rB^2}+\frac{C'B'}{2CB^2}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}+\frac{B'D_5'}{4B^2D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{C''}{CB}-\frac{D_4''}{2BD_4}-\frac{D_5''}{2BD_5}=\Lambda$$

$$R_{22}: \quad \frac{1}{r^2C}-\frac{1}{r^2B}-\frac{2C'}{rCB}+\frac{B'}{2rB^2}+\frac{C'B'}{4CB^2}-\frac{D_4'}{2rBD_4}-\frac{C'D_4'}{4CBD_4}-\frac{D_5'}{2rBD_5}-\frac{C'D_5'}{4CBD_5}-\frac{C''}{2CB}=\Lambda$$

$$R_{44}: \quad \frac{\dot{a}^2}{a^2}+\frac{\ddot{a}}{a}-\frac{D_4'}{rBD_4}-\frac{C'D_4'}{2CBD_4}+\frac{B'D_4'}{4B^2D_4}+\frac{(D_4')^2}{4BD_4^2}-\frac{D_4'D_5'}{4BD_4D_5}-\frac{D_4''}{2BD_4}=\Lambda$$

$$R_{55}: \quad \frac{\dot{a}^2}{a^2}+\frac{\ddot{a}}{a}-\frac{D_5'}{rBD_5}-\frac{C'D_5'}{2CBD_5}+\frac{B'D_5'}{4B^2D_5}-\frac{D_4'D_5'}{4BD_4D_5}+\frac{(D_5')^2}{4BD_5^2}-\frac{D_5''}{2BD_5}=\Lambda$$

From R_{01}, we see that

$$D_5(r) = \frac{K}{D_4}$$ where K is a constant of integration.

We can simplify D_5 by 'rolling K into p_5' as explained above in Section aii:

$$D_5(r) = \pm\frac{1}{D_4}.$$

From R_{00} we get

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\beta$$ where α and β are constants of integration.

That makes R_{00} and R_{01} identities, and the remaining equations become

$$R_{11}: \quad -\frac{2C'}{rCB} + \frac{(C')^2}{2C^2B} + \frac{B'}{rB^2} + \frac{C'B'}{2CB^2} - \frac{(D_4')^2}{2BD_4^2} - \frac{C''}{CB} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2C} - \frac{1}{r^2B} - \frac{2C'}{rCB} + \frac{B'}{2rB^2} + \frac{C'B'}{4CB^2} - \frac{C''}{2CB} = \Lambda$$

$$R_{44}: \quad \frac{e^{2\sqrt{2}t\sqrt{\Lambda}}\alpha^2\Lambda}{\left(e^{\sqrt{2}t\sqrt{\Lambda}}\alpha+\beta\right)^2} + \frac{\beta^2\Lambda}{\left(e^{\sqrt{2}t\sqrt{\Lambda}}\alpha+\beta\right)^2} - \frac{D_4'}{rBD_4} - \frac{C'D_4'}{2CBD_4} + \frac{B'D_4'}{4B^2D_4} + \frac{(D_4')^2}{2BD_4^2} - \frac{D_4''}{2BD_4} = \Lambda$$

$$R_{55}: \quad \frac{e^{2\sqrt{2}t\sqrt{\Lambda}}\alpha^2\Lambda}{\left(e^{\sqrt{2}t\sqrt{\Lambda}}\alpha+\beta\right)^2} + \frac{\beta^2\Lambda}{\left(e^{\sqrt{2}t\sqrt{\Lambda}}\alpha+\beta\right)^2} + \frac{D_4'}{rBD_4} + \frac{C'D_4'}{2CBD_4} - \frac{B'D_4'}{4B^2D_4} - \frac{(D_4')^2}{2BD_4^2} + \frac{D_4''}{2BD_4} = \Lambda$$

Observe that

$$R_{44}+R_{55}: \quad \frac{2e^{2\sqrt{2}t\sqrt{\Lambda}}\alpha^2\Lambda}{\left(e^{\sqrt{2}t\sqrt{\Lambda}}\alpha+\beta\right)^2} + \frac{2\beta^2\Lambda}{\left(e^{\sqrt{2}t\sqrt{\Lambda}}\alpha+\beta\right)^2}$$

$$= \frac{2\left(e^{2\sqrt{2}t\sqrt{\Lambda}}\alpha^2+\beta^2\right)\Lambda}{\left(e^{\sqrt{2}t\sqrt{\Lambda}}\alpha+\beta\right)^2} = 2\Lambda$$

$R_{44}+R_{55}$ is an identity if either $\alpha=0$ or $\beta=0$:

$$R_{44}+R_{55}: \quad 2\Lambda = 2\Lambda$$

Since $R_{44}+R_{55}=2\Lambda$, the two equations are dependent and we can declare R_{55} redundant.

Observe that $\alpha=0$ or $\beta=0$ is equivalent to

$$a(t) = e^{\pm\frac{t\sqrt{\Lambda}}{\sqrt{2}}}\alpha$$

This formula for $a(t)$ simplifies R_{44}, and the equations are now

$$R_{11}: \quad -\frac{2C'}{rCB} + \frac{(C')^2}{2C^2B} + \frac{B'}{rB^2} + \frac{C'B'}{2CB^2} - \frac{(D_4')^2}{2BD_4^2} - \frac{C''}{CB} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{C''}{2\,C\,B} = \Lambda$$

$$R_{44}: \quad \Lambda - \frac{D_4'}{r\,B\,D_4} - \frac{C'\,D_4'}{2\,C\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{2\,B\,D_4^2} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

We could algebraically solve $R_{11}-2R_{22}$ for B, but the solution is complex and makes the remaining equations unwieldy. Rather, we can solve that equation for D_4:

$$R_{11}-2R_{22}: \quad -\frac{2}{r^2\,C} + \frac{2}{r^2\,B} + \frac{2\,C'}{r\,C\,B} + \frac{(C')^2}{2\,C^2\,B} - \frac{(D_4')^2}{2\,B\,D_4^2} = -\Lambda$$

$$D_4(r) = e^{\int_1^r -\frac{\pm\sqrt{4\,C^2-4\,C\,B+2\,C^2\,B\,\Lambda\,\rho^2+4\,C\,\rho\,C'+\rho^2\,(C')^2}}{C\,\rho}\,d\rho}\,\gamma \quad \text{where } \gamma \text{ is a constant of integration.}$$

As with D_5, we can roll γ into the radius p_4:

$$D_4(r) = \pm e^{\int_1^r -\frac{\pm\sqrt{4\,C^2-4\,C\,B+2\,C^2\,B\,\Lambda\,\rho^2+4\,C\,\rho\,C'+\rho^2\,(C')^2}}{C\,\rho}\,d\rho}$$

Since $R_{11}-2R_{22}=-\Lambda$, R_{11} is redundant. This leaves only two equations, R_{22} and R_{44}, in the two unknowns B and C. The integral cancels out (i.e., disappears) in both equations but R_{44} is too lengthy to display here. R_{22} becomes

$$R_{22}: \quad \frac{1}{r^2\,C} - \frac{1}{r^2\,B} - \frac{2\,C'}{r\,C\,B} + \frac{B'}{2\,r\,B^2} + \frac{C'\,B'}{4\,C\,B^2} - \frac{C''}{2\,C\,B} = \Lambda$$

Solving this equation for B yields

$$B(r) = -\frac{r^2\,\Lambda\,(2\,C + r\,C')^2}{2 - 4\,r^2\,C\,\Lambda - \zeta\,\Lambda + 2\,r^4\,C^2\,\Lambda^2}, \quad \text{where } \zeta \text{ is a c onstant of integration.}$$

Plugging this in, the remaining equation is also an identity:

$$R_{44}: \quad \Lambda = \Lambda$$

We have finally found another exact solution to the Λ field equations:

$$a(t) = e^{\pm\frac{t\,\sqrt{\Lambda}}{\sqrt{2}}}\,\alpha \quad \text{where } \alpha \text{ is a free constant,}$$

$$A(t,r) = c^2,$$

$$B(t,r) = -\frac{r^2\,\Lambda\,(2\,C[r] + r\,C'[r])^2}{2 - \zeta\,\Lambda - 4\,r^2\,\Lambda\,C[r] + 2\,r^4\,\Lambda^2\,C[r]^2} \quad \text{where } \zeta \text{ is a free constant,}$$

$$\mathbb{C}(t,r) = C[r], \text{ a free parameter,}$$

$$\mathbb{D}_4(t,r) = \pm a(t)^2 D_4(r) = \pm a(t)^2 e^{\int_1^r -\frac{\sqrt{-2+\zeta\,\Lambda}\,(2\,C[\rho]+\rho\,C'[\rho])}{\rho\,C[\rho]\,\sqrt{-2+\zeta\,\Lambda+4\,\Lambda\,\rho^2\,C[\rho]-2\,\Lambda^2\,\rho^4\,C[\rho]^2}}\,d\rho},$$

$$\mathbb{D}_5(t,r) = \pm\frac{a[t]^2}{D_4[r]} = \pm a[t]^2 e^{-\int_1^r -\frac{\sqrt{-2+\zeta\,\Lambda}\,(2\,C[\rho]+\rho\,C'[\rho])}{\rho\,C[\rho]\,\sqrt{-2+\zeta\,\Lambda+4\,\Lambda\,\rho^2\,C[\rho]-2\,\Lambda^2\,\rho^4\,C[\rho]^2}}\,d\rho},$$

and line element

$$ds^2 = c^2\, dt^2 + \frac{r^2 \Lambda\, (2\, C[r] + r\, C'[r])^2}{2 - \zeta \Lambda - 4\, r^2 \Lambda\, C[r] + 2\, r^4 \Lambda^2\, C[r]^2}\, dr^2 - r^2\, C[r]\, d\theta^2 - r^2\, \mathrm{Sin}[\theta]^2\, C[r]\, d\phi^2$$

$$- e^{\sqrt{2}\, t\, \sqrt{\Lambda} + \int_1^r -\frac{\sqrt{-2+\zeta\Lambda}\, (2\, C[\rho] + \rho\, C'[\rho])}{\rho\, C[\rho]\, \sqrt{-2+\zeta\Lambda+4\Lambda\rho^2 C[\rho] - 2\Lambda^2 \rho^4 C[\rho]^2}}\, d\rho}\, \alpha^2\, p_4^2\, d\psi_4^2$$

$$- e^{\sqrt{2}\, t\, \sqrt{\Lambda} - \int_1^r -\frac{\sqrt{-2+\zeta\Lambda}\, (2\, C[\rho] + \rho\, C'[\rho])}{\rho\, C[\rho]\, \sqrt{-2+\zeta\Lambda+4\Lambda\rho^2 C[\rho] - 2\Lambda^2 \rho^4 C[\rho]^2}}\, d\rho}\, \alpha^2\, p_5^2\, d\psi_5^2$$

For example, if we set ζ=0 and $C[r]$=1, then

$$B(t,r) = -\frac{2\, r^2 \Lambda}{\left(-1 + r^2 \Lambda\right)^2}$$

and $\quad D_4(r) = \pm \dfrac{r^2\, (1 - \Lambda)}{1 - r^2 \Lambda}$

bii: B(t,r)= B(r), C(t,r) = 1

We next try a subcase, Case bii, of Case b. Starting fresh, the field equations become

$$R_{00}: \quad \frac{2\, c^2\, \ddot{a}}{a\, A} - \frac{c^2\, \dot{a}\, \dot{A}}{a\, A^2} - \frac{A'}{A\, r\, B} + \frac{(A')^2}{4\, A^2\, B} + \frac{A'\, B'}{4\, A\, B^2} - \frac{A'\, D_4'}{4\, A\, B\, D_4} - \frac{A'\, D_5'}{4\, A\, B\, D_5} - \frac{A''}{2\, A\, B} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a}\, A'}{a\, A} + \frac{\dot{a}\, D_4'}{2\, a\, D_4} + \frac{\dot{a}\, D_5'}{2\, a\, D_5} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4\, A^2\, B} + \frac{B'}{r\, B^2} + \frac{A'\, B'}{4\, A\, B^2} + \frac{B'\, D_4'}{4\, B^2\, D_4} + \frac{(D_4')^2}{4\, B\, D_4^2} + \frac{B'\, D_5'}{4\, B^2\, D_5} + \frac{(D_5')^2}{4\, B\, D_5^2} - \frac{A''}{2\, A\, B} - \frac{D_4''}{2\, B\, D_4} - \frac{D_5''}{2\, B\, D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{r^2} - \frac{1}{r^2\, B} - \frac{A'}{2\, A\, r\, B} + \frac{B'}{2\, r\, B^2} - \frac{D_4'}{2\, r\, B\, D_4} - \frac{D_5'}{2\, r\, B\, D_5} = \Lambda$$

$$R_{44}: \quad \frac{c^2\, \dot{a}^2}{a^2\, A} + \frac{c^2\, \ddot{a}}{a\, A} - \frac{c^2\, \dot{a}\, \dot{A}}{2\, a\, A^2} - \frac{D_4'}{r\, B\, D_4} - \frac{A'\, D_4'}{4\, A\, B\, D_4} + \frac{B'\, D_4'}{4\, B^2\, D_4} + \frac{(D_4')^2}{4\, B\, D_4^2} - \frac{D_4'\, D_5'}{4\, B\, D_4\, D_5} - \frac{D_4''}{2\, B\, D_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2\, \dot{a}^2}{a^2\, A} + \frac{c^2\, \ddot{a}}{a\, A} - \frac{c^2\, \dot{a}\, \dot{A}}{2\, a\, A^2} - \frac{D_5'}{r\, B\, D_5} - \frac{A'\, D_5'}{4\, A\, B\, D_5} + \frac{B'\, D_5'}{4\, B^2\, D_5} - \frac{D_4'\, D_5'}{4\, B\, D_4\, D_5} + \frac{(D_5')^2}{4\, B\, D_5^2} - \frac{D_5''}{2\, B\, D_5} = \Lambda$$

Like Case b, we can simplify R_{01} by factoring out $\dfrac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \quad \frac{A'}{A} - \frac{D_4'}{2\, D_4} - \frac{D_5'}{2\, D_5} = 0$$

$A(t,r) = \mathcal{A}\, \sqrt{D_4}\, \sqrt{D_5}$ where $\mathcal{A}(t)$ is a new unknown.

The equations now become

$$R_{00}: \quad \frac{2\, c^2\, \ddot{a}}{a\, \mathcal{A}\, \sqrt{D_4}\, \sqrt{D_5}} - \frac{c^2\, \dot{a}\, \dot{\mathcal{A}}}{a\, \mathcal{A}^2\, \sqrt{D_4}\, \sqrt{D_5}} - \frac{D_4'}{2\, r\, B\, D_4} + \frac{B'\, D_4'}{8\, B^2\, D_4} +$$

$$\frac{(D_4')^2}{16\, B\, D_4^2} - \frac{D_5'}{2\, r\, B\, D_5} + \frac{B'\, D_5'}{8\, B^2\, D_5} - \frac{3\, D_4'\, D_5'}{8\, B\, D_4\, D_5} + \frac{(D_5')^2}{16\, B\, D_5^2} - \frac{D_4''}{4\, B\, D_4} - \frac{D_5''}{4\, B\, D_5} = \Lambda$$

R_{11}: $$\frac{B'}{r B^2} + \frac{3 B' D_4'}{8 B^2 D_4} + \frac{7 (D_4')^2}{16 B D_4^2} + \frac{3 B' D_5'}{8 B^2 D_5} - \frac{D_4' D_5'}{8 B D_4 D_5} + \frac{7 (D_5')^2}{16 B D_5^2} - \frac{3 D_4''}{4 B D_4} - \frac{3 D_5''}{4 B D_5} = \Lambda$$

R_{22}: $$\frac{1}{r^2} - \frac{1}{r^2 B} + \frac{B'}{2 r B^2} - \frac{3 D_4'}{4 r B D_4} - \frac{3 D_5'}{4 r B D_5} = \Lambda$$

R_{44}:

$$\frac{c^2 \dot{a}^2}{a^2 \mathcal{A} \sqrt{D_4} \sqrt{D_5}} + \frac{c^2 \ddot{a}}{a \mathcal{A} \sqrt{D_4} \sqrt{D_5}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \sqrt{D_4} \sqrt{D_5}} - \frac{D_4'}{r B D_4} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{8 B D_4^2} - \frac{3 D_4' D_5'}{8 B D_4 D_5} - \frac{D_4''}{2 B D_4} = \Lambda$$

R_{55}:

$$\frac{c^2 \dot{a}^2}{a^2 \mathcal{A} \sqrt{D_4} \sqrt{D_5}} + \frac{c^2 \ddot{a}}{a \mathcal{A} \sqrt{D_4} \sqrt{D_5}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2 \sqrt{D_4} \sqrt{D_5}} - \frac{D_5'}{r B D_5} + \frac{B' D_5'}{4 B^2 D_5} - \frac{3 D_4' D_5'}{8 B D_4 D_5} + \frac{(D_5')^2}{8 B D_5^2} - \frac{D_5''}{2 B D_5} = \Lambda$$

From $2R_{00}-R_{44}-R_{55}$ we can solve for $\mathcal{A}(t)$:

$2R_{00}-R_{44}-R_{55}$: $$-\frac{2 c^2 \dot{a}^2}{a^2 \mathcal{A} \sqrt{D_4} \sqrt{D_5}} + \frac{2 c^2 \ddot{a}}{a \mathcal{A} \sqrt{D_4} \sqrt{D_5}} - \frac{c^2 \dot{a} \dot{\mathcal{A}}}{a \mathcal{A}^2 \sqrt{D_4} \sqrt{D_5}} = 0$$

$\Rightarrow$ $2R_{00}-R_{44}-R_{55}$: $$-2 c^2 \mathcal{A} \dot{a}^2 + 2 a c^2 \mathcal{A} \ddot{a} - a c^2 \dot{a} \dot{\mathcal{A}} = 0$$

$$\mathcal{A}(t) = \frac{k \dot{a}[t]^2}{a[t]^2}$$ where k is a constant of integration.

With this formula, R_{00}, R_{44}, and R_{55} are dependent equations, so we can choose to designate R_{44} as redundant.

The remaining equations become:

R_{00}:

$$\frac{2 c^2}{k \sqrt{D_4} \sqrt{D_5}} - \frac{D_4'}{2 r B D_4} + \frac{B' D_4'}{8 B^2 D_4} + \frac{(D_4')^2}{16 B D_4^2} - \frac{D_5'}{2 r B D_5} + \frac{B' D_5'}{8 B^2 D_5} - \frac{3 D_4' D_5'}{8 B D_4 D_5} + \frac{(D_5')^2}{16 B D_5^2} - \frac{D_4''}{4 B D_4} - \frac{D_5''}{4 B D_5} = \Lambda$$

R_{11}: $$\frac{B'}{r B^2} + \frac{3 B' D_4'}{8 B^2 D_4} + \frac{7 (D_4')^2}{16 B D_4^2} + \frac{3 B' D_5'}{8 B^2 D_5} - \frac{D_4' D_5'}{8 B D_4 D_5} + \frac{7 (D_5')^2}{16 B D_5^2} - \frac{3 D_4''}{4 B D_4} - \frac{3 D_5''}{4 B D_5} = \Lambda$$

R_{22}: $$\frac{1}{r^2} - \frac{1}{r^2 B} + \frac{B'}{2 r B^2} - \frac{3 D_4'}{4 r B D_4} - \frac{3 D_5'}{4 r B D_5} = \Lambda$$

R_{55}: $$\frac{2 c^2}{k \sqrt{D_4} \sqrt{D_5}} - \frac{D_5'}{r B D_5} + \frac{B' D_5'}{4 B^2 D_5} - \frac{3 D_4' D_5'}{8 B D_4 D_5} + \frac{(D_5')^2}{8 B D_5^2} - \frac{D_5''}{2 B D_5} = \Lambda$$

We can solve R_{22} for D_5:

$$D_5(r) = e^{\int_1^r \frac{-4 B D_4 + 4 B^2 D_4 - 4 B^2 \Lambda \rho^2 D_4 + 2 \rho D_4 B' - 3 B \rho D_4'}{3 B \rho D_4} d\rho} \alpha$$ where α is a constant of integration.

We can roll up α into the radius p_5 (as explained in Section aii):

$$D_5(r) = e^{\int_1^r \frac{-4 B D_4 + 4 B^2 D_4 - 4 B^2 \Lambda \rho^2 D_4 + 2 \rho D_4 B' - 3 B \rho D_4'}{3 B \rho D_4} d\rho}$$

R_{22} is eliminated. R_{00} and R_{55} are messy and involve an integral expression. But, R_{11} is short enough to show here:

$$R_{11}: \quad \frac{19}{9 r^2} - \frac{14}{9 r^2 B} - \frac{5 B}{9 r^2} - \frac{\Lambda}{9} + \frac{10 B \Lambda}{9} - \frac{5}{9} r^2 B \Lambda^2 + \frac{19 B'}{18 r B^2} - \frac{19 B'}{18 r B} + \frac{19 r \Lambda B'}{18 B} + \frac{11 (B')^2}{18 B^3} + \frac{2 D_4'}{3 r D_4} - \frac{2 D_4'}{3 r B D_4} - \frac{2 r \Lambda D_4'}{3 D_4} + \frac{B' D_4'}{3 B^2 D_4} - \frac{(D_4')^2}{2 B D_4^2} - \frac{B''}{2 B^2} = \Lambda$$

We can try to solve R_{11} for both the remaining unknowns, B and D_4. Interestingly, the integral disappears in $R_{00}-R_{55}$, so we can also try solving it and also $3R_{00}+R_{11}-3R_{55}$:

$$R_{00}-R_{55}: \quad -\frac{1}{3 r^2} + \frac{B}{3 r^2} - \frac{\Lambda}{3} - \frac{2 B \Lambda}{3} + \frac{1}{3} r^2 B \Lambda^2 + \frac{B'}{6 r B^2} + \frac{B'}{2 r B} - \frac{r \Lambda B'}{2 B} - \frac{(B')^2}{6 B^3} - \frac{D_4'}{2 r D_4} - \frac{D_4'}{2 r B D_4} + \frac{r \Lambda D_4'}{2 D_4} + \frac{(D_4')^2}{2 B D_4^2} + \frac{B''}{6 B^2} - \frac{D_4''}{2 B D_4} = 0$$

$$3R_{00}+R_{11}-3R_{55} \quad : \frac{10}{9 r^2} - \frac{14}{9 r^2 B} + \frac{4 B}{9 r^2} - \frac{10 \Lambda}{9} - \frac{8 B \Lambda}{9} + \frac{4}{9} r^2 B \Lambda^2 + \frac{14 B'}{9 r B^2} + \frac{4 B'}{9 r B} - \frac{4 r \Lambda B'}{9 B} + \frac{(B')^2}{9 B^3} - \frac{5 D_4'}{6 r D_4} - \frac{13 D_4'}{6 r B D_4} + \frac{5 r \Lambda D_4'}{6 D_4} + \frac{B' D_4'}{3 B^2 D_4} + \frac{(D_4')^2}{B D_4^2} - \frac{3 D_4''}{2 B D_4} = \Lambda$$

However, we are unable to solve any of these equations in such a way that the remaining equations are subsequently solvable.

This case, too, is too difficult for me to solve, so we proceed to **Case biii**: $B(r) = 1$, $C(r) = 1$.

biii: B(t,r)=1, C(t,r)=1

Starting fresh again, the field equations become

$$R_{00}: \quad \frac{2 c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{a A^2} - \frac{A'}{A r} + \frac{(A')^2}{4 A^2} - \frac{A' D_4'}{4 A D_4} - \frac{A' D_5'}{4 A D_5} - \frac{A''}{2 A} = \Lambda$$

$$R_{01}: \quad -\frac{\dot{a} A'}{a A} + \frac{\dot{a} D_4'}{2 a D_4} + \frac{\dot{a} D_5'}{2 a D_5} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4 A^2} + \frac{(D_4')^2}{4 D_4^2} + \frac{(D_5')^2}{4 D_5^2} - \frac{A''}{2 A} - \frac{D_4''}{2 D_4} - \frac{D_5''}{2 D_5} = \Lambda$$

$$R_{22}: \quad -\frac{A'}{2 A r} - \frac{D_4'}{2 r D_4} - \frac{D_5'}{2 r D_5} = \Lambda$$

$$R_{44}: \quad \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{D_4'}{r D_4} - \frac{A' D_4'}{4 A D_4} + \frac{(D_4')^2}{4 D_4^2} - \frac{D_4' D_5'}{4 D_4 D_5} - \frac{D_4''}{2 D_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{D_5'}{r D_5} - \frac{A' D_5'}{4 A D_5} - \frac{D_4' D_5'}{4 D_4 D_5} + \frac{(D_5')^2}{4 D_5^2} - \frac{D_5''}{2 D_5} = \Lambda$$

Again, like Case b, we can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \quad \frac{A'}{A} - \frac{\mathcal{D}_4'}{2\,\mathcal{D}_4} - \frac{\mathcal{D}_5'}{2\,\mathcal{D}_5} = 0$$

$A(t,r) = \mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}$ where $\mathcal{A}(t)$ is a new unknown.

The equations now become

$$R_{00}: \quad \frac{2\,c^2\,\ddot{a}}{a\,\mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{\mathcal{D}_4'}{2\,r\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{16\,\mathcal{D}_4^2} - \frac{\mathcal{D}_5'}{2\,r\,\mathcal{D}_5} - \frac{3\,\mathcal{D}_4'\,\mathcal{D}_5'}{8\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{16\,\mathcal{D}_5^2} - \frac{\mathcal{D}_4''}{4\,\mathcal{D}_4} - \frac{\mathcal{D}_5''}{4\,\mathcal{D}_5} = \Lambda$$

$$R_{11}: \quad \frac{7\,(\mathcal{D}_4')^2}{16\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{8\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{7\,(\mathcal{D}_5')^2}{16\,\mathcal{D}_5^2} - \frac{3\,\mathcal{D}_4''}{4\,\mathcal{D}_4} - \frac{3\,\mathcal{D}_5''}{4\,\mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{3\,\mathcal{D}_4'}{4\,r\,\mathcal{D}_4} - \frac{3\,\mathcal{D}_5'}{4\,r\,\mathcal{D}_5} = \Lambda$$

$$R_{44}: \quad \frac{c^2\,\dot{a}^2}{a^2\,\mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{\mathcal{D}_4'}{r\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{8\,\mathcal{D}_4^2} - \frac{3\,\mathcal{D}_4'\,\mathcal{D}_5'}{8\,\mathcal{D}_4\,\mathcal{D}_5} - \frac{\mathcal{D}_4''}{2\,\mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{c^2\,\dot{a}^2}{a^2\,\mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} + \frac{c^2\,\ddot{a}}{a\,\mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{\mathcal{D}_5'}{r\,\mathcal{D}_5} - \frac{3\,\mathcal{D}_4'\,\mathcal{D}_5'}{8\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{8\,\mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\,\mathcal{D}_5} = \Lambda$$

From $2R_{00}-R_{44}-R_{55}$ we can solve for $\mathcal{A}(t)$:

$$2R_{00}-R_{44}-R_{55}: \quad -\frac{2\,c^2\,\dot{a}^2}{a^2\,\mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} + \frac{2\,c^2\,\ddot{a}}{a\,\mathcal{A}\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,\mathcal{A}^2\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} = 0$$

$$\Rightarrow \quad 2R_{00}-R_{44}-R_{55}: \quad -2\,c^2\,\mathcal{A}\,\dot{a}^2 + 2\,a\,c^2\,\mathcal{A}\,\ddot{a} - a\,c^2\,\dot{a}\,\dot{\mathcal{A}} = 0$$

$\mathcal{A}(t) = \frac{k\,\dot{a}[t]^2}{a[t]^2}$ where k is a constant of integration.

With this formula, R_{00}, R_{44}, and R_{55} are dependent equations, so we can choose to designate R_{44} as redundant.

The remaining equations become:

$$R_{00}: \quad \frac{2\,c^2}{k\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{\mathcal{D}_4'}{2\,r\,\mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{16\,\mathcal{D}_4^2} - \frac{\mathcal{D}_5'}{2\,r\,\mathcal{D}_5} - \frac{3\,\mathcal{D}_4'\,\mathcal{D}_5'}{8\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{16\,\mathcal{D}_5^2} - \frac{\mathcal{D}_4''}{4\,\mathcal{D}_4} - \frac{\mathcal{D}_5''}{4\,\mathcal{D}_5} = \Lambda$$

$$R_{11}: \quad \frac{7\,(\mathcal{D}_4')^2}{16\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{8\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{7\,(\mathcal{D}_5')^2}{16\,\mathcal{D}_5^2} - \frac{3\,\mathcal{D}_4''}{4\,\mathcal{D}_4} - \frac{3\,\mathcal{D}_5''}{4\,\mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{3\,\mathcal{D}_4'}{4\,r\,\mathcal{D}_4} - \frac{3\,\mathcal{D}_5'}{4\,r\,\mathcal{D}_5} = \Lambda$$

$$R_{55}: \quad \frac{2\,c^2}{k\sqrt{\mathcal{D}_4}\sqrt{\mathcal{D}_5}} - \frac{\mathcal{D}_5'}{r\,\mathcal{D}_5} - \frac{3\,\mathcal{D}_4'\,\mathcal{D}_5'}{8\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{8\,\mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\,\mathcal{D}_5} = \Lambda$$

We can solve R_{22} for $\mathcal{D}_5$:

$$\mathcal{D}_5(r) = \frac{e^{-\frac{2 r^2 \Lambda}{3}} \alpha}{\mathcal{D}_4}$$ where α is a constant of integration.

R_{22} is eliminated. The equations now become:

$$R_{00}: \quad \frac{2 c^2 e^{\frac{r^2 \Lambda}{3}}}{k \sqrt{\alpha}} + \Lambda - \frac{r^2 \Lambda^2}{3} = \Lambda$$

$$R_{11}: \quad \Lambda - \frac{5 r^2 \Lambda^2}{9} - \frac{2 r \Lambda \mathcal{D}_4'}{3 \mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{2 \mathcal{D}_4^2} = \Lambda$$

$$R_{55}: \quad \frac{2 c^2 e^{\frac{r^2 \Lambda}{3}}}{k \sqrt{\alpha}} + 2 \Lambda - \frac{2 r^2 \Lambda^2}{3} + \frac{\mathcal{D}_4'}{r \mathcal{D}_4} - \frac{r \Lambda \mathcal{D}_4'}{2 \mathcal{D}_4} - \frac{(\mathcal{D}_4')^2}{2 \mathcal{D}_4^2} + \frac{\mathcal{D}_4''}{2 \mathcal{D}_4} = \Lambda$$

Rearranging R_{00}, we get:

$$6 c^2 e^{\frac{r^2 \Lambda}{3}} = k r^2 \sqrt{\alpha} \Lambda^2,$$

which has no solution for a constant Λ.

We conclude that this case has no possible solution.

c: B(t,r)=B(r), $\mathbb{C}$(t,r)=a(t)^2

The field equations become

$$R_{00}: \quad \frac{4 c^2 \ddot{a}}{a A} - \frac{2 c^2 \dot{a} \dot{A}}{a A^2} - \frac{A'}{A r B} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' \mathcal{D}_4'}{4 A B \mathcal{D}_4} - \frac{A' \mathcal{D}_5'}{4 A B \mathcal{D}_5} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad \frac{2 \dot{a}}{a r} - \frac{2 \dot{a} A'}{a A} + \frac{\dot{a} \mathcal{D}_4'}{2 a \mathcal{D}_4} + \frac{\dot{a} \mathcal{D}_5'}{2 a \mathcal{D}_5} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4 A^2 B} + \frac{B'}{r B^2} + \frac{A' B'}{4 A B^2} + \frac{B' \mathcal{D}_4'}{4 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 B \mathcal{D}_4^2} + \frac{B' \mathcal{D}_5'}{4 B^2 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 B \mathcal{D}_5^2} - \frac{A''}{2 A B} - \frac{\mathcal{D}_4''}{2 B \mathcal{D}_4} - \frac{\mathcal{D}_5''}{2 B \mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{1}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{A'}{2 A r B} + \frac{B'}{2 r B^2} - \frac{\mathcal{D}_4'}{2 r B \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 r B \mathcal{D}_5} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{\mathcal{D}_4'}{r B \mathcal{D}_4} - \frac{A' \mathcal{D}_4'}{4 A B \mathcal{D}_4} + \frac{B' \mathcal{D}_4'}{4 B^2 \mathcal{D}_4} + \frac{(\mathcal{D}_4')^2}{4 B \mathcal{D}_4^2} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 B \mathcal{D}_4 \mathcal{D}_5} - \frac{\mathcal{D}_4''}{2 B \mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{c^2 \dot{a} \dot{A}}{2 a A^2} - \frac{\mathcal{D}_5'}{r B \mathcal{D}_5} - \frac{A' \mathcal{D}_5'}{4 A B \mathcal{D}_5} + \frac{B' \mathcal{D}_5'}{4 B^2 \mathcal{D}_5} - \frac{\mathcal{D}_4' \mathcal{D}_5'}{4 B \mathcal{D}_4 \mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4 B \mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2 B \mathcal{D}_5} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}: \quad -\frac{2}{r} + \frac{2 A'}{A} - \frac{\mathcal{D}_4'}{2 \mathcal{D}_4} - \frac{\mathcal{D}_5'}{2 \mathcal{D}_5} = 0$$

$A(t,r) = r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}$ where $\mathcal{A}(t)$ is a new unknown.

This eliminates R_{01}, and the remaining 5 equations become

$$R_{00}:\quad -\frac{3}{4\,r^2\,B}+\frac{4\,c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}-\frac{2\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,r\,\mathcal{A}^2\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}+\frac{B'}{4\,r\,B^2}-\frac{5\,\mathcal{D}_4'}{8\,r\,B\,\mathcal{D}_4}+$$
$$\frac{B'\,\mathcal{D}_4'}{16\,B^2\,\mathcal{D}_4}+\frac{3\,(\mathcal{D}_4')^2}{64\,B\,\mathcal{D}_4^2}-\frac{5\,\mathcal{D}_5'}{8\,r\,B\,\mathcal{D}_5}+\frac{B'\,\mathcal{D}_5'}{16\,B^2\,\mathcal{D}_5}-\frac{5\,\mathcal{D}_4'\,\mathcal{D}_5'}{32\,B\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{3\,(\mathcal{D}_5')^2}{64\,B\,\mathcal{D}_5^2}-\frac{\mathcal{D}_4''}{8\,B\,\mathcal{D}_4}-\frac{\mathcal{D}_5''}{8\,B\,\mathcal{D}_5}=\Lambda$$

$$R_{11}:\quad \frac{1}{4\,r^2\,B}+\frac{5\,B'}{4\,r\,B^2}-\frac{\mathcal{D}_4'}{8\,r\,B\,\mathcal{D}_4}+\frac{5\,B'\,\mathcal{D}_4'}{16\,B^2\,\mathcal{D}_4}+\frac{23\,(\mathcal{D}_4')^2}{64\,B\,\mathcal{D}_4^2}-$$
$$\frac{\mathcal{D}_5'}{8\,r\,B\,\mathcal{D}_5}+\frac{5\,B'\,\mathcal{D}_5'}{16\,B^2\,\mathcal{D}_5}-\frac{\mathcal{D}_4'\,\mathcal{D}_5'}{32\,B\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{23\,(\mathcal{D}_5')^2}{64\,B\,\mathcal{D}_5^2}-\frac{5\,\mathcal{D}_4''}{8\,B\,\mathcal{D}_4}-\frac{5\,\mathcal{D}_5''}{8\,B\,\mathcal{D}_5}=\Lambda$$

R_{22}:

$$\frac{1}{a^2\,r^2}-\frac{3}{2\,r^2\,B}+\frac{3\,c^2\,\dot{a}^2}{a^2\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}+\frac{c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}-\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,r\,\mathcal{A}^2\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}+\frac{B'}{2\,r\,B^2}-\frac{5\,\mathcal{D}_4'}{8\,r\,B\,\mathcal{D}_4}-\frac{5\,\mathcal{D}_5'}{8\,r\,B\,\mathcal{D}_5}=\Lambda$$

$$R_{44}:\quad \frac{3\,c^2\,\dot{a}^2}{a^2\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}+\frac{c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}-$$
$$\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,r\,\mathcal{A}^2\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}-\frac{5\,\mathcal{D}_4'}{4\,r\,B\,\mathcal{D}_4}+\frac{B'\,\mathcal{D}_4'}{4\,B^2\,\mathcal{D}_4}+\frac{3\,(\mathcal{D}_4')^2}{16\,B\,\mathcal{D}_4^2}-\frac{5\,\mathcal{D}_4'\,\mathcal{D}_5'}{16\,B\,\mathcal{D}_4\,\mathcal{D}_5}-\frac{\mathcal{D}_4''}{2\,B\,\mathcal{D}_4}=\Lambda$$

$$R_{55}:\quad \frac{3\,c^2\,\dot{a}^2}{a^2\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}+\frac{c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}-$$
$$\frac{c^2\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,r\,\mathcal{A}^2\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}-\frac{5\,\mathcal{D}_5'}{4\,r\,B\,\mathcal{D}_5}+\frac{B'\,\mathcal{D}_5'}{4\,B^2\,\mathcal{D}_5}-\frac{5\,\mathcal{D}_4'\,\mathcal{D}_5'}{16\,B\,\mathcal{D}_4\,\mathcal{D}_5}+\frac{3\,(\mathcal{D}_5')^2}{16\,B\,\mathcal{D}_5^2}-\frac{\mathcal{D}_5''}{2\,B\,\mathcal{D}_5}=\Lambda$$

Now we can solve $4R_{00}-2R_{22}-R_{44}-R_{55}$ for $\mathcal{A}$:

$$4R_{00}-2R_{22}-R_{44}-R_{55}:\quad -\frac{2}{a^2\,r^2}-\frac{12\,c^2\,\dot{a}^2}{a^2\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}+\frac{12\,c^2\,\ddot{a}}{a\,r\,\mathcal{A}\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}-\frac{6\,c^2\,\dot{a}\,\dot{\mathcal{A}}}{a\,r\,\mathcal{A}^2\,\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}=0$$

$\mathcal{A}(t) = \dfrac{6\,c^2\,r\,\dot{a}^2}{6\,a^2\,c^2\,r\,\alpha-\mathcal{D}_4^{1/4}\,\mathcal{D}_5^{1/4}}$ where α is a constant of integration;

This formula for $\mathcal{A}(t)$ makes R_{00}, R_{22}, R_{44},
and R_{55} dependent, so we are free to declare R_{00} redundant.

Also, in order for $\mathcal{A}(t)$ to be strictly a function of t, $\sqrt[4]{\mathcal{D}_4\,\mathcal{D}_5}$ must be proportional to r:

$\sqrt[4]{\mathcal{D}_4\,\mathcal{D}_5} = K\,r$ where K is the proportionality constant.

Solving for this yields

$$\Rightarrow\quad \mathcal{D}_5(r) = \frac{r^4\,K^4}{\mathcal{D}_4}$$

Plugging $\mathcal{A}(t)$ and $D_5(r)$ back in, the equations become:

$$R_{11}: \quad -\frac{2}{r^2 B} + \frac{5 B'}{2 r B^2} + \frac{2 D_4'}{r B D_4} - \frac{(D_4')^2}{2 B D_4^2} = \Lambda$$

$$R_{22}: \quad \frac{1}{2 a^2 r^2} - \frac{4}{r^2 B} + \frac{4 c^2 \alpha}{r^2 K} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{44}: \quad -\frac{7 c^2 \alpha}{r (6 a^2 c^2 r \alpha - r K)} + \frac{24 a^2 c^4 \alpha^2}{r K (6 a^2 c^2 r \alpha - r K)} + \frac{K}{2 a^2 r (6 a^2 c^2 r \alpha - r K)} - \frac{15 a^2 c^2 \alpha D_4'}{B (6 a^2 c^2 r \alpha - r K) D_4} + \frac{5 K D_4'}{2 B (6 a^2 c^2 r \alpha - r K) D_4} + \frac{3 a^2 c^2 r \alpha B' D_4'}{2 B^2 (6 a^2 c^2 r \alpha - r K) D_4} - \frac{r K B' D_4'}{4 B^2 (6 a^2 c^2 r \alpha - r K) D_4} + \frac{3 a^2 c^2 r \alpha (D_4')^2}{B (6 a^2 c^2 r \alpha - r K) D_4^2} - \frac{r K (D_4')^2}{2 B (6 a^2 c^2 r \alpha - r K) D_4^2} - \frac{3 a^2 c^2 r \alpha D_4''}{B (6 a^2 c^2 r \alpha - r K) D_4} + \frac{r K D_4''}{2 B (6 a^2 c^2 r \alpha - r K) D_4} = \Lambda$$

$$R_{55}: \quad -\frac{7 c^2 \alpha}{r (6 a^2 c^2 r \alpha - r K)} - \frac{48 a^2 c^2 \alpha}{r B (6 a^2 c^2 r \alpha - r K)} + \frac{24 a^2 c^4 \alpha^2}{r K (6 a^2 c^2 r \alpha - r K)} + \frac{K}{2 a^2 r (6 a^2 c^2 r \alpha - r K)} + \frac{8 K}{r B (6 a^2 c^2 r \alpha - r K)} + \frac{6 a^2 c^2 \alpha B'}{B^2 (6 a^2 c^2 r \alpha - r K)} - \frac{K B'}{B^2 (6 a^2 c^2 r \alpha - r K)} + \frac{15 a^2 c^2 \alpha D_4'}{B (6 a^2 c^2 r \alpha - r K) D_4} - \frac{5 K D_4'}{2 B (6 a^2 c^2 r \alpha - r K) D_4} - \frac{3 a^2 c^2 r \alpha B' D_4'}{2 B^2 (6 a^2 c^2 r \alpha - r K) D_4} + \frac{r K B' D_4'}{4 B^2 (6 a^2 c^2 r \alpha - r K) D_4} - \frac{3 a^2 c^2 r \alpha (D_4')^2}{B (6 a^2 c^2 r \alpha - r K) D_4^2} + \frac{r K (D_4')^2}{2 B (6 a^2 c^2 r \alpha - r K) D_4^2} + \frac{3 a^2 c^2 r \alpha D_4''}{B (6 a^2 c^2 r \alpha - r K) D_4} - \frac{r K D_4''}{2 B (6 a^2 c^2 r \alpha - r K) D_4} = \Lambda$$

Solving R_{22} for B yields

$$B(r) = -\frac{40 r^8 K a[t]^2}{-5 r^8 K - 40 c^2 r^8 \alpha a[t]^2 - 40 \beta K a[t]^2 + 8 r^{10} K \Lambda a[t]^2}$$

where β is a constant of integration.

In order for $B(r)$ to be strictly a function of r, the $a(t)^2$ term in the numerator must cancel with the denominator. The $-5r^8K$ term prevents this unless $K=0$. But, then both $B(r)=0$ and $D_5(r)=0$, contradictions.

This case has no possible solution.

We next re-do this case, but with various additional assumptions. By making the assumptions upfront, before the ODEs have been solved, it is possible to obtain distinct solutions. We try a couple of these subcases even though all of the subcases in the 5D notebook had no possible solution.

Iβ: A(t,r) = A(r)

II: A(t,r) = c^2

c_Iβ: B(t,r)=B(r), C(t,r)=a(t)^2, A(t,r)=A(r)

Starting fresh, the field equations are identical to those in Case c except that all instances of A(t,r) are replaced by A(r):

$$R_{00}: \quad \frac{4 c^2 \ddot{a}}{a A} - \frac{A'}{r A B} + \frac{(A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A' D_4'}{4 A B D_4} - \frac{A' D_5'}{4 A B D_5} - \frac{A''}{2 A B} = \Lambda$$

$$R_{01}: \quad \frac{2 \dot{a}}{a r} - \frac{2 \dot{a} A'}{a A} + \frac{\dot{a} D_4'}{2 a D_4} + \frac{\dot{a} D_5'}{2 a D_5} = 0$$

$$R_{11}: \quad \frac{(A')^2}{4 A^2 B} + \frac{B'}{r B^2} + \frac{A' B'}{4 A B^2} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{4 B D_4^2} + \frac{B' D_5'}{4 B^2 D_5} + \frac{(D_5')^2}{4 B D_5^2} - \frac{A''}{2 A B} - \frac{D_4''}{2 B D_4} - \frac{D_5''}{2 B D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} - \frac{1}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{A'}{2 r A B} + \frac{B'}{2 r B^2} - \frac{D_4'}{2 r B D_4} - \frac{D_5'}{2 r B D_5} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D_4'}{r B D_4} - \frac{A' D_4'}{4 A B D_4} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{4 B D_4^2} - \frac{D_4' D_5'}{4 B D_4 D_5} - \frac{D_4''}{2 B D_4} = \Lambda$$

$$R_{55}: \quad \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D_5'}{r B D_5} - \frac{A' D_5'}{4 A B D_5} + \frac{B' D_5'}{4 B^2 D_5} - \frac{D_4' D_5'}{4 B D_4 D_5} + \frac{(D_5')^2}{4 B D_5^2} - \frac{D_5''}{2 B D_5} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for D_5:

$$R_{01}: \quad -\frac{2}{r} + \frac{2 A'}{A} - \frac{D_4'}{2 D_4} - \frac{D_5'}{2 D_5} = 0$$

$$D_5(r) = \frac{k A^4}{r^4 D_4}$$ where k is a constant of integration

This eliminates R_{01}, and the remaining 4 equations become

$$R_{00}: \quad \frac{4 c^2 \ddot{a}}{a A} - \frac{3 (A')^2}{4 A^2 B} + \frac{A' B'}{4 A B^2} - \frac{A''}{2 A B} = \Lambda$$

$$R_{11}: \quad -\frac{6}{r^2 B} + \frac{8 A'}{r A B} - \frac{7 (A')^2}{4 A^2 B} + \frac{5 A' B'}{4 A B^2} - \frac{2 D_4'}{r B D_4} + \frac{2 A' D_4'}{A B D_4} - \frac{(D_4')^2}{2 B D_4^2} - \frac{5 A''}{2 A B} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{1}{r^2 B} + \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{5 A'}{2 r A B} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{44}: \quad \frac{3 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{5 A' D_4'}{4 A B D_4} + \frac{B' D_4'}{4 B^2 D_4} + \frac{(D_4')^2}{2 B D_4^2} - \frac{D_4''}{2 B D_4} = \Lambda$$

R_{55}:

$$-\frac{2}{r^2 B}+\frac{3 c^2 \dot{a}^2}{a^2 A}+\frac{c^2 \ddot{a}}{a A}+\frac{5 A'}{r A B}-\frac{3 (A')^2}{A^2 B}-\frac{B'}{r B^2}+\frac{A' B'}{A B^2}+\frac{5 A' D_4'}{4 A B D_4}-\frac{B' D_4'}{4 B^2 D_4}-\frac{(D_4')^2}{2 B D_4^2}-\frac{2 A''}{A B}+\frac{D_4''}{2 B D_4} = \Lambda$$

Now we can algebraically solve for A:

$$4R_{00}-2R_{22}-R_{44}-R_{55}: \quad -\frac{2}{a^2 r^2}-\frac{12 c^2 \dot{a}^2}{a^2 A}+\frac{12 c^2 \ddot{a}}{a A} = 0$$

$$\Rightarrow \quad A(r) = -6 c^2 r^2 \left(\dot{a}[t]^2 - a[t]\, \ddot{a}[t]\right)$$

In order for $A(r)$ to be strictly a function of r,
the expression in the parentheses must equal a constant, say K:

Solving for this yields

$$a(t) = -\frac{\sqrt{K}\, \mathrm{Sinh}[t\,\alpha+\beta]}{\alpha} \quad \text{where } \alpha,\ \beta, \text{ and K are constants}$$

$$\Rightarrow \quad A(r) = -6 c^2 K r^2$$

With this expression for A, (but ignoring $a(t)$ for
now), $4R_{00}-2R_{22}-R_{44}-R_{55}=0$; these equations are dependent and we
are free to designate R_{55} as redundant. The equations then become:

R_{00}: $$-\frac{4}{r^2 B}-\frac{2 \ddot{a}}{3 a K r^2}+\frac{B'}{2 r B^2} = \Lambda$$

R_{11}: $$-\frac{2}{r^2 B}+\frac{5 B'}{2 r B^2}+\frac{2 D_4'}{r B D_4}-\frac{(D_4')^2}{2 B D_4^2} = \Lambda$$

R_{22}: $$\frac{1}{a^2 r^2}-\frac{4}{r^2 B}-\frac{\dot{a}^2}{2 a^2 K r^2}-\frac{\ddot{a}}{6 a K r^2}+\frac{B'}{2 r B^2} = \Lambda$$

R_{44}: $$-\frac{\dot{a}^2}{2 a^2 K r^2}-\frac{\ddot{a}}{6 a K r^2}-\frac{5 D_4'}{2 r B D_4}+\frac{B' D_4'}{4 B^2 D_4}+\frac{(D_4')^2}{2 B D_4^2}-\frac{D_4''}{2 B D_4} = \Lambda$$

$R_{00}-R_{22}$ has no solution:

$$R_{00}-R_{22}: \quad -\frac{1}{r^2 a[t]^2}+\frac{\dot{a}[t]^2}{2 K r^2 a[t]^2}-\frac{\ddot{a}[t]}{2 K r^2 a[t]} = 0$$

Plugging in $a(t)$ makes this

$$R_{00}-R_{22}: \quad -\frac{\alpha^2 \mathrm{Csch}[t\,\alpha+\beta]^2}{2 K r^2} = 0$$

Since $\mathrm{Csch}^2(t)>0$ for all t, we must have

$$\alpha = 0$$

which makes $a(t)$ undefined.

This case has no possible solution.

c_II: B(t,r)=B(r), C(t,r)=a(t)^2, A(t,r)=c^2

Starting fresh, the field equations are:

$$R_{00}: \quad \frac{4\,\ddot{a}}{a} = \Lambda$$

$$R_{01}: \quad \frac{2\,\dot{a}}{a\,r} + \frac{\dot{a}\,D_4'}{2\,a\,D_4} + \frac{\dot{a}\,D_5'}{2\,a\,D_5} = 0$$

$$R_{11}: \quad \frac{B'}{r\,B^2} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{4\,B\,D_4^2} + \frac{B'\,D_5'}{4\,B^2\,D_5} + \frac{(D_5')^2}{4\,B\,D_5^2} - \frac{D_4''}{2\,B\,D_4} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} - \frac{1}{r^2\,B} + \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{B'}{2\,r\,B^2} - \frac{D_4'}{2\,r\,B\,D_4} - \frac{D_5'}{2\,r\,B\,D_5} = \Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_4'}{r\,B\,D_4} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{4\,B\,D_4^2} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_5'}{r\,B\,D_5} + \frac{B'\,D_5'}{4\,B^2\,D_5} - \frac{D_4'\,D_5'}{4\,B\,D_4\,D_5} + \frac{(D_5')^2}{4\,B\,D_5^2} - \frac{D_5''}{2\,B\,D_5} = \Lambda$$

Solving R_{00} yields

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{2}}\,\alpha + e^{-\frac{t\sqrt{\Lambda}}{2}}\,\beta.$$

We don't actually use this formula. We will
continue to work with generic a(t), noting that R_{00} is an identity.

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for D_5:

$$R_{01}: \quad -\frac{2}{r} - \frac{D_4'}{2\,D_4} - \frac{D_5'}{2\,D_5} = 0$$

$$D_5(r) = \frac{K}{r^4\,D_4} \quad \text{where } K \text{ is a constant of integration}$$

R_{00} and R_{01} are identities, and the remaining 4 equations become

$$R_{11}: \quad -\frac{6}{r^2\,B} - \frac{2\,D_4'}{r\,B\,D_4} - \frac{(D_4')^2}{2\,B\,D_4^2} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} + \frac{1}{r^2\,B} + \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{B'}{2\,r\,B^2} = \Lambda$$

$$R_{44}: \quad \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{B'\,D_4'}{4\,B^2\,D_4} + \frac{(D_4')^2}{2\,B\,D_4^2} - \frac{D_4''}{2\,B\,D_4} = \Lambda$$

$$R_{55}: \quad -\frac{2}{r^2\,B} + \frac{3\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{B'}{r\,B^2} - \frac{B'\,D_4'}{4\,B^2\,D_4} - \frac{(D_4')^2}{2\,B\,D_4^2} + \frac{D_4''}{2\,B\,D_4} = \Lambda$$

We can solve R_{11} for D_4:

$$D_4(r) = e^{\int_1^r \frac{i\left(2i+\sqrt{2}\sqrt{4+\Lambda\rho^2 B[\rho]}\right)}{\rho} d\rho}\gamma \quad \text{where } \gamma \text{ is a constant of integration.}$$

R_{11} has also become an identity and the integral in $D_4(r)$ cancels out in the remaining equations:

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{1}{r^2 B} + \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{B'}{2 r B^2} = \Lambda$$

$$R_{44}: \quad -\frac{1}{r^2 B} + \frac{2i}{r^2 B\sqrt{2+\frac{1}{2}r^2 B\Lambda}} + \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{B'}{2 r B^2} + \frac{i B'}{r B^2\sqrt{2+\frac{1}{2}r^2 B\Lambda}} = \Lambda$$

$$R_{55}: \quad -\frac{1}{r^2 B} - \frac{2i}{r^2 B\sqrt{2+\frac{1}{2}r^2 B\Lambda}} + \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{B'}{2 r B^2} - \frac{i B'}{r B^2\sqrt{2+\frac{1}{2}r^2 B\Lambda}} = \Lambda$$

$R_{44}-R_{55}$ can be solved for B:

$$R_{44}-R_{55}: \quad \frac{4i}{r^2 B\sqrt{2+\frac{1}{2}r^2 B\Lambda}} + \frac{2i B'}{r B^2\sqrt{2+\frac{1}{2}r^2 B\Lambda}}$$

$$= \frac{2i\,(2B + r B')}{r^2 B^2\sqrt{2+\frac{1}{2}r^2 B\Lambda}} = 0$$

$$B(r) = \frac{k}{r^2} \quad \text{where k is a constant of integration.}$$

Using this formula, the integral in the formula for $D_4(r)$ can be evaluated:

$$D_4(r) = e^{\int_1^r \frac{i\left(2i+\sqrt{2}\sqrt{4+k\Lambda}\right)}{\rho} d\rho}\gamma$$

$$= r^{-2+i\sqrt{8+2k\Lambda}}\gamma \quad \text{where } r\neq 1 \text{ (or else } D_4(r) = \text{constant).}$$

The equations now become:

$$R_{22}: \quad \frac{1}{a^2 r^2} + \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

$$R_{44}: \quad \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

$$R_{55}: \quad \frac{3\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

$$R_{22}-R_{44}: \quad \frac{1}{r^2 a[t]^2} = 0$$

This contradiction shows that this case has no solution.

d: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=\mathbb{C}(r)$

The field equations become

$$R_{00}:\quad \frac{3c^2\ddot{a}}{aA}-\frac{3c^2\dot{a}\dot{A}}{2aA^2}-\frac{A'}{a^2Ar}+\frac{(A')^2}{4a^2A^2}-\frac{A'\mathbb{C}'}{2a^2A\mathbb{C}}-\frac{A'\mathbb{D}_4'}{4a^2A\mathbb{D}_4}-\frac{A'\mathbb{D}_5'}{4a^2A\mathbb{D}_5}-\frac{A''}{2a^2A}=\Lambda$$

$$R_{01}:\quad -\frac{2\dot{a}}{ar}-\frac{\dot{a}A'}{aA}-\frac{\dot{a}\mathbb{C}'}{a\mathbb{C}}=0$$

$$R_{11}:\quad \frac{2c^2\dot{a}^2}{a^2A}+\frac{c^2\ddot{a}}{aA}-\frac{c^2\dot{a}\dot{A}}{2aA^2}+\frac{(A')^2}{4a^2A^2}-\frac{2\mathbb{C}'}{a^2r\mathbb{C}}+$$
$$\frac{(\mathbb{C}')^2}{2a^2\mathbb{C}^2}+\frac{(\mathbb{D}_4')^2}{4a^2\mathbb{D}_4^2}+\frac{(\mathbb{D}_5')^2}{4a^2\mathbb{D}_5^2}-\frac{A''}{2a^2A}-\frac{\mathbb{C}''}{a^2\mathbb{C}}-\frac{\mathbb{D}_4''}{2a^2\mathbb{D}_4}-\frac{\mathbb{D}_5''}{2a^2\mathbb{D}_5}=\Lambda$$

$$R_{22}:\quad -\frac{1}{a^2r^2}+\frac{1}{r^2\mathbb{C}}-\frac{A'}{2a^2Ar}-\frac{2\mathbb{C}'}{a^2r\mathbb{C}}-\frac{A'\mathbb{C}'}{4a^2A\mathbb{C}}-\frac{\mathbb{D}_4'}{2a^2r\mathbb{D}_4}-\frac{\mathbb{C}'\mathbb{D}_4'}{4a^2\mathbb{C}\mathbb{D}_4}-\frac{\mathbb{D}_5'}{2a^2r\mathbb{D}_5}-\frac{\mathbb{C}'\mathbb{D}_5'}{4a^2\mathbb{C}\mathbb{D}_5}-\frac{\mathbb{C}''}{2a^2\mathbb{C}}=\Lambda$$

$$R_{44}:\quad \frac{2c^2\dot{a}^2}{a^2A}+\frac{c^2\ddot{a}}{aA}-\frac{c^2\dot{a}\dot{A}}{2aA^2}-\frac{\mathbb{D}_4'}{a^2r\mathbb{D}_4}-\frac{A'\mathbb{D}_4'}{4a^2A\mathbb{D}_4}-\frac{\mathbb{C}'\mathbb{D}_4'}{2a^2\mathbb{C}\mathbb{D}_4}+\frac{(\mathbb{D}_4')^2}{4a^2\mathbb{D}_4^2}-\frac{\mathbb{D}_4'\mathbb{D}_5'}{4a^2\mathbb{D}_4\mathbb{D}_5}-\frac{\mathbb{D}_4''}{2a^2\mathbb{D}_4}=\Lambda$$

$$R_{55}:\quad \frac{2c^2\dot{a}^2}{a^2A}+\frac{c^2\ddot{a}}{aA}-\frac{c^2\dot{a}\dot{A}}{2aA^2}-\frac{\mathbb{D}_5'}{a^2r\mathbb{D}_5}-\frac{A'\mathbb{D}_5'}{4a^2A\mathbb{D}_5}-\frac{\mathbb{C}'\mathbb{D}_5'}{2a^2\mathbb{C}\mathbb{D}_5}-\frac{\mathbb{D}_4'\mathbb{D}_5'}{4a^2\mathbb{D}_4\mathbb{D}_5}+\frac{(\mathbb{D}_5')^2}{4a^2\mathbb{D}_5^2}-\frac{\mathbb{D}_5''}{2a^2\mathbb{D}_5}=\Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$, and then solving for A:

$$R_{01}:\quad \frac{2}{r}+\frac{A'}{A}+\frac{\mathbb{C}'}{\mathbb{C}}=0$$

$$A(t,r)=\frac{\mathcal{A}[t]}{r^2\mathbb{C}[r]}$$ where $\mathcal{A}(t)$ is a new unknown.

This eliminates R_{01}, and the remaining equations become

$$R_{00}:\quad \frac{3c^2r^2\mathbb{C}\ddot{a}}{a\mathcal{A}}-\frac{3c^2r^2\mathbb{C}\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2}+\frac{\mathbb{C}'}{a^2r\mathbb{C}}-\frac{(\mathbb{C}')^2}{4a^2\mathbb{C}^2}+\frac{\mathbb{D}_4'}{2a^2r\mathbb{D}_4}+\frac{\mathbb{C}'\mathbb{D}_4'}{4a^2\mathbb{C}\mathbb{D}_4}+\frac{\mathbb{D}_5'}{2a^2r\mathbb{D}_5}+\frac{\mathbb{C}'\mathbb{D}_5'}{4a^2\mathbb{C}\mathbb{D}_5}+\frac{\mathbb{C}''}{2a^2\mathbb{C}}=\Lambda$$

$$R_{11}:\quad -\frac{2}{a^2r^2}+\frac{2c^2r^2\mathbb{C}\dot{a}^2}{a^2\mathcal{A}}+\frac{c^2r^2\mathbb{C}\ddot{a}}{a\mathcal{A}}-\frac{c^2r^2\mathbb{C}\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2}-$$
$$\frac{3\mathbb{C}'}{a^2r\mathbb{C}}-\frac{(\mathbb{C}')^2}{4a^2\mathbb{C}^2}+\frac{(\mathbb{D}_4')^2}{4a^2\mathbb{D}_4^2}+\frac{(\mathbb{D}_5')^2}{4a^2\mathbb{D}_5^2}-\frac{\mathbb{C}''}{2a^2\mathbb{C}}-\frac{\mathbb{D}_4''}{2a^2\mathbb{D}_4}-\frac{\mathbb{D}_5''}{2a^2\mathbb{D}_5}=\Lambda$$

$$R_{22}:\quad \frac{1}{r^2\mathbb{C}}-\frac{\mathbb{C}'}{a^2r\mathbb{C}}+\frac{(\mathbb{C}')^2}{4a^2\mathbb{C}^2}-\frac{\mathbb{D}_4'}{2a^2r\mathbb{D}_4}-\frac{\mathbb{C}'\mathbb{D}_4'}{4a^2\mathbb{C}\mathbb{D}_4}-\frac{\mathbb{D}_5'}{2a^2r\mathbb{D}_5}-\frac{\mathbb{C}'\mathbb{D}_5'}{4a^2\mathbb{C}\mathbb{D}_5}-\frac{\mathbb{C}''}{2a^2\mathbb{C}}=\Lambda$$

$$R_{44}:\quad \frac{2c^2r^2\mathbb{C}\dot{a}^2}{a^2\mathcal{A}}+\frac{c^2r^2\mathbb{C}\ddot{a}}{a\mathcal{A}}-\frac{c^2r^2\mathbb{C}\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2}-\frac{\mathbb{D}_4'}{2a^2r\mathbb{D}_4}-\frac{\mathbb{C}'\mathbb{D}_4'}{4a^2\mathbb{C}\mathbb{D}_4}+\frac{(\mathbb{D}_4')^2}{4a^2\mathbb{D}_4^2}-\frac{\mathbb{D}_4'\mathbb{D}_5'}{4a^2\mathbb{D}_4\mathbb{D}_5}-\frac{\mathbb{D}_4''}{2a^2\mathbb{D}_4}=\Lambda$$

$$R_{55}:\quad \frac{2c^2r^2\mathbb{C}\dot{a}^2}{a^2\mathcal{A}}+\frac{c^2r^2\mathbb{C}\ddot{a}}{a\mathcal{A}}-\frac{c^2r^2\mathbb{C}\dot{a}\dot{\mathcal{A}}}{2a\mathcal{A}^2}-\frac{\mathbb{D}_5'}{2a^2r\mathbb{D}_5}-\frac{\mathbb{C}'\mathbb{D}_5'}{4a^2\mathbb{C}\mathbb{D}_5}-\frac{\mathbb{D}_4'\mathbb{D}_5'}{4a^2\mathbb{D}_4\mathbb{D}_5}+\frac{(\mathbb{D}_5')^2}{4a^2\mathbb{D}_5^2}-\frac{\mathbb{D}_5''}{2a^2\mathbb{D}_5}=\Lambda$$

We can solve $R_{00}+R_{22}$ for $\mathcal{A}$:

$$R_{00}+R_{22}: \quad \frac{1}{r^2\,C} + \frac{3\,c^2\,r^2\,C\,\ddot{a}}{a\,\mathcal{A}} - \frac{3\,c^2\,r^2\,C\,\dot{a}\,\dot{\mathcal{A}}}{2\,a\,\mathcal{A}^2} = 2\,\Lambda$$

$$\mathcal{A}(t) = \frac{3\,c^2\,r^4\,C[r]^2\,\dot{a}[t]^2}{-a[t]^2 + 2\,r^2\,\Lambda\,a[t]^2\,C[r] + 3\,c^2\,r^4\,\kappa\,C[r]^2}$$

$$= \frac{3\,c^2\,\dot{a}[t]^2}{3\,c^2\,\kappa + a[t]^2\left(-\frac{1}{r^4\,C[r]^2} + \frac{2\,\Lambda}{r^2\,C[r]}\right)}$$

where κ is a constant of integration.

In order for $\mathcal{A}(t)$ not to be a function of r, the term in parentheses must be a constant.

$$\Rightarrow \quad C(r) = \frac{K}{r^2} \quad \text{for some proportionality constant } K,$$

$$\Rightarrow \quad \mathcal{A}(t) = \frac{3\,c^2\,K^2\,\dot{a}[t]^2}{3\,c^2\,\kappa\,K^2 - a[t]^2 + 2\,K\,\Lambda\,a[t]^2}.$$

Moreover, this makes A(t,r) strictly a function of t:

$$A(t,r) = \frac{\mathcal{A}[t]}{r^2\,C[r]} = \frac{\mathcal{A}(t)}{K}.$$

The equations now become

$$R_{00}: \quad -\frac{1}{K} + 2\,\Lambda = \Lambda$$

$$R_{11}: \quad -\frac{1}{K} + \frac{2\,c^2\,\kappa\,K}{a^2} + 2\,\Lambda + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,\mathcal{D}_5^2} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} - \frac{\mathcal{D}_5''}{2\,a^2\,\mathcal{D}_5} = \Lambda$$

$$R_{22}: \quad \frac{1}{K} = \Lambda$$

$$R_{44}: \quad -\frac{1}{K} + \frac{2\,c^2\,\kappa\,K}{a^2} + 2\,\Lambda + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{D}_4\,\mathcal{D}_5} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad -\frac{1}{K} + \frac{2\,c^2\,\kappa\,K}{a^2} + 2\,\Lambda - \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{D}_4\,\mathcal{D}_5} + \frac{(\mathcal{D}_5')^2}{4\,a^2\,\mathcal{D}_5^2} - \frac{\mathcal{D}_5''}{2\,a^2\,\mathcal{D}_5} = \Lambda$$

From R_{22} we get

$$\Lambda \neq 0 \quad \text{and} \quad K = \frac{1}{\Lambda}.$$

This modifies A(t,r) again:

$$A(t,r) = \Lambda\,\mathcal{A}(t) = (\Lambda)\left(\frac{3\,c^2\,\dot{a}[t]^2}{3\,c^2\,\kappa - \Lambda^2\,a[t]^2 + 2\,\Lambda^2\,a[t]^2}\right) = \frac{3\,c^2\,\Lambda\,\dot{a}[t]^2}{3\,c^2\,\kappa + \Lambda^2\,a[t]^2}.$$

R_{00} and R_{22} have become identities, and the other equations become

$$R_{11}: \quad \frac{2\,c^2\,\kappa}{a^2\,\Lambda} + \frac{(D_4')^2}{4\,a^2\,D_4^2} + \frac{(D_5')^2}{4\,a^2\,D_5^2} - \frac{D_4''}{2\,a^2\,D_4} - \frac{D_5''}{2\,a^2\,D_5} = 0$$

$$R_{44}: \quad \frac{2\,c^2\,\kappa}{a^2\,\Lambda} + \frac{(D_4')^2}{4\,a^2\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,D_4} = 0$$

$$R_{55}: \quad \frac{2\,c^2\,\kappa}{a^2\,\Lambda} - \frac{D_4'\,D_5'}{4\,a^2\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,D_5^2} - \frac{D_5''}{2\,a^2\,D_5} = 0$$

We can solve $R_{11}-R_{44}-R_{55}$ for D_5:

$$R_{11}-R_{44}-R_{55}: \quad -\frac{2\,c^2\,\kappa}{a^2\,\Lambda} - \Lambda + \frac{D_4'\,D_5'}{2\,a^2\,D_4\,D_5} = -\Lambda$$

$$D_5(r) = \gamma\, e^{4c^2\kappa \int_1^r \frac{D_4}{D_4'}\, d\rho}$$ where γ is a constant of integration.

Since $g_{55}=-a(t)^2 D_5(r) p_5{}^2$, we can roll the positive part of γ into the radius p_5:

$$D_5(r) = \pm e^{4c^2\kappa \int_1^r \frac{D_4}{D_4'}\, d\rho}$$

Plugging this in, the integral cancels out:

$$R_{11}: \quad -\frac{4\,c^4\,\kappa^2\,D_4^2}{a^2\,\Lambda^2\,(D_4')^2} + \frac{(D_4')^2}{4\,a^2\,D_4^2} - \frac{D_4''}{2\,a^2\,D_4} + \frac{2\,c^2\,\kappa\,D_4\,D_4''}{a^2\,\Lambda\,(D_4')^2} = 0$$

$$R_{44}: \quad \frac{c^2\,\kappa}{a^2\,\Lambda} + \frac{(D_4')^2}{4\,a^2\,D_4^2} - \frac{D_4''}{2\,a^2\,D_4} = 0$$

$$R_{55}: \quad -\frac{c^2\,\kappa}{a^2\,\Lambda} - \frac{4\,c^4\,\kappa^2\,D_4^2}{a^2\,\Lambda^2\,(D_4')^2} + \frac{2\,c^2\,\kappa\,D_4\,D_4''}{a^2\,\Lambda\,(D_4')^2} = 0$$

We can solve $R_{11}-R_{44}$ for the remaining unknown, $D_4(r)$:

$$R_{11}-R_{44}: \quad -\frac{c^2\,\kappa}{a^2\,\Lambda} - \frac{4\,c^4\,\kappa^2\,D_4^2}{a^2\,\Lambda^2\,(D_4')^2} + \frac{2\,c^2\,\kappa\,D_4\,D_4''}{a^2\,\Lambda\,(D_4')^2} = 0$$

$$D_4(r) = \chi\,\mathrm{Cosh}\left[\frac{c\,\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2$$ where χ and ω are constants of integration.

Since $g_{44}=-a(t)^2 D_4(r) p_4{}^2$, we can roll the positive part of χ into the radius p_4:

$$D_4(r) = \pm\mathrm{Cosh}\left[\frac{c\,\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2$$

$$\Rightarrow \quad D_5(r) = \pm e^{\int_1^r \frac{2\,c\,\sqrt{\kappa}\,\mathrm{Coth}\left[\frac{c\,\sqrt{\kappa}\,(\rho-\omega)}{\sqrt{\Lambda}}\right]}{\sqrt{\Lambda}}\, d\rho} = \pm e^{2\left(-\mathrm{Log}\left[\mathrm{Sinh}\left[\frac{c\,\sqrt{\kappa}\,(-1+\omega)}{\sqrt{\Lambda}}\right]\right]+\mathrm{Log}\left[\mathrm{Sinh}\left[\frac{c\,\sqrt{\kappa}\,(-r+\omega)}{\sqrt{\Lambda}}\right]\right]\right)}$$

$$= \pm\mathrm{Csch}\left[\frac{c\,\sqrt{\kappa}\,(1-\omega)}{\sqrt{\Lambda}}\right]^2 \mathrm{Sinh}\left[\frac{c\,\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2$$

The Csch term is a constant that can be either positive or negative. Thus, we can roll the entire Csch term into the radius p_5:

$$\Rightarrow \quad \mathbb{D}_5(r) = \pm \text{Sinh}\left[\frac{c\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2.$$

Plugging $\mathbb{D}_4$ and $\mathbb{D}_5$ in, the remaining equations are satisfied:

R_{11}: $\Lambda = \Lambda$

R_{44}: $\Lambda = \Lambda$

R_{55}: $\Lambda = \Lambda$

All of the equations are satisfied, so we have an exact solution to the $\Lambda \neq 0$ field equations:

$a(t)$ is a free parameter,

$$A(t,r) = \frac{3c^2\Lambda\,\dot{a}[t]^2}{3c^2\kappa + \Lambda^2 a[t]^2} \quad \text{where } \kappa \text{ is a free constant,}$$

$$B(t,r) = a[t]^2,$$

$$\mathbb{C}(t,r) = \frac{1}{r^2\Lambda} \quad \text{where } \Lambda \neq 0,$$

$$\mathbb{D}_4(t,r) = \pm a(t)^2 \mathbb{D}_4(r) = \pm a[t]^2 \,\text{Cosh}\left[\frac{c\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2$$

where ω is a free constant,

$$\mathbb{D}_5(t,r) = \pm a(t)^2 \mathbb{D}_5(r) = \pm a[t]^2 \,\text{Sinh}\left[\frac{c\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2,$$

and line element

$$ds^2 = \frac{3c^2\Lambda\,\dot{a}[t]^2}{3c^2\kappa + \Lambda^2 a[t]^2}\,dt^2 - a[t]^2\,dr^2 - \frac{1}{\Lambda}\,d\theta^2 - \frac{\text{Sin}[\theta]^2}{\Lambda}\,d\phi^2$$
$$- a[t]^2\,\text{Cosh}\left[\frac{c\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2 p_4^2\,d\psi_4^2$$
$$- a[t]^2\,\text{Sinh}\left[\frac{c\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2 p_5^2\,d\psi_5^2$$

d-II: Solve $A(t,r)=c^2$ for $a(t)$

Solving $\dfrac{3c^2\Lambda\,\dot{a}[t]^2}{3c^2\kappa + \Lambda^2 a[t]^2} = c^2$ yields

$$a(t) = \pm\frac{\sqrt{3}\,c\sqrt{\kappa}\,\text{Sinh}\left[\frac{\sqrt{\Lambda}\,(t-\upsilon)}{\sqrt{3}}\right]}{\Lambda} \quad \text{where } \upsilon \text{ is a free constant.}$$

dIβ: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=\mathbb{C}(r)$, $A(t,r)=A(r)$

As usual, when we re-do a case with an additional assumption, we could obtain another solution. We are re-doing Case d with the assumption that $\mathcal{A}(t) = 1$; i.e., $A(t,r) = \mathcal{A}(t)\, A(r) = A(r)$.

Starting fresh, the field equations become

$$R_{00}:\quad \frac{3 c^2 \ddot{a}}{a A} - \frac{A'}{a^2 r A} - \frac{C' A'}{2 a^2 C A} + \frac{(A')^2}{4 a^2 A^2} - \frac{A' D_4'}{4 a^2 A D_4} - \frac{A' D_5'}{4 a^2 A D_5} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{01}:\quad -\frac{2 \dot{a}}{a r} - \frac{\dot{a} C'}{a C} - \frac{\dot{a} A'}{a A} = 0$$

$$R_{11}:\quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{2 C'}{a^2 r C} + \frac{(C')^2}{2 a^2 C^2} + \frac{(A')^2}{4 a^2 A^2} + \frac{(D_4')^2}{4 a^2 D_4^2} + \frac{(D_5')^2}{4 a^2 D_5^2} - \frac{C''}{a^2 C} - \frac{A''}{2 a^2 A} - \frac{D_4''}{2 a^2 D_4} - \frac{D_5''}{2 a^2 D_5} = \Lambda$$

$$R_{22}:\quad -\frac{1}{a^2 r^2} + \frac{1}{r^2 C} - \frac{2 C'}{a^2 r C} - \frac{A'}{2 a^2 r A} - \frac{C' A'}{4 a^2 C A} - \frac{D_4'}{2 a^2 r D_4} - \frac{C' D_4'}{4 a^2 C D_4} - \frac{D_5'}{2 a^2 r D_5} - \frac{C' D_5'}{4 a^2 C D_5} - \frac{C''}{2 a^2 C} = \Lambda$$

$$R_{44}:\quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D_4'}{a^2 r D_4} - \frac{C' D_4'}{2 a^2 C D_4} - \frac{A' D_4'}{4 a^2 A D_4} + \frac{(D_4')^2}{4 a^2 D_4^2} - \frac{D_4' D_5'}{4 a^2 D_4 D_5} - \frac{D_4''}{2 a^2 D_4} = \Lambda$$

$$R_{55}:\quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{D_5'}{a^2 r D_5} - \frac{C' D_5'}{2 a^2 C D_5} - \frac{A' D_5'}{4 a^2 A D_5} - \frac{D_4' D_5'}{4 a^2 D_4 D_5} + \frac{(D_5')^2}{4 a^2 D_5^2} - \frac{D_5''}{2 a^2 D_5} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$:

$$R_{01}:\quad \frac{2}{r} + \frac{C'}{C} + \frac{A'}{A} = 0$$

We can solve R_{01} either for A in terms of C or C in terms of A. We will do the latter:

$$C[r] = \frac{\sqrt{3}\, c \sqrt{\kappa}}{r^2 \Lambda A[r]}$$ where κ is a constant of integration and $\Lambda \neq 0$.

This eliminates R_{01}, and the remaining equations become

$$R_{00}:\quad \frac{3 c^2 \ddot{a}}{a A} + \frac{3 (A')^2}{4 a^2 A^2} - \frac{A' D_4'}{4 a^2 A D_4} - \frac{A' D_5'}{4 a^2 A D_5} - \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{11}:\quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} - \frac{5 (A')^2}{4 a^2 A^2} + \frac{(D_4')^2}{4 a^2 D_4^2} + \frac{(D_5')^2}{4 a^2 D_5^2} + \frac{A''}{2 a^2 A} - \frac{D_4''}{2 a^2 D_4} - \frac{D_5''}{2 a^2 D_5} = \Lambda$$

$$R_{22}:\quad \frac{A \Lambda}{\sqrt{3}\, c \sqrt{\kappa}} - \frac{3 (A')^2}{4 a^2 A^2} + \frac{A' D_4'}{4 a^2 A D_4} + \frac{A' D_5'}{4 a^2 A D_5} + \frac{A''}{2 a^2 A} = \Lambda$$

$$R_{44}:\quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} + \frac{A' D_4'}{4 a^2 A D_4} + \frac{(D_4')^2}{4 a^2 D_4^2} - \frac{D_4' D_5'}{4 a^2 D_4 D_5} - \frac{D_4''}{2 a^2 D_4} = \Lambda$$

$$R_{55}:\quad \frac{2 c^2 \dot{a}^2}{a^2 A} + \frac{c^2 \ddot{a}}{a A} + \frac{A' D_5'}{4 a^2 A D_5} - \frac{D_4' D_5'}{4 a^2 D_4 D_5} + \frac{(D_5')^2}{4 a^2 D_5^2} - \frac{D_5''}{2 a^2 D_5} = \Lambda$$

We can algebraically solve $R_{00}+R_{22}$ for A:

$$R_{00}+R_{22}: \quad \frac{A\,\Lambda}{\sqrt{3}\,c\,\sqrt{\kappa}} + \frac{3\,c^2\,\ddot{a}}{a\,A} = 2\,\Lambda$$

$$\Rightarrow \quad A(r) = \frac{\sqrt{3}\left(c\,\sqrt{\kappa}\,\Lambda\,a[t] + \sqrt{c^2\,\sqrt{\kappa}\,\Lambda\,a[t]\left(\sqrt{\kappa}\,\Lambda\,a[t] - \sqrt{3}\,c\,\ddot{a}[t]\right)}\right)}{\Lambda\,a[t]}.$$

Observe that there are no r terms on the RHS,
so if the t-terms cancel out, that will make $A(r)$ a constant.

In order for the t-terms to cancel out, the $a(t)$
terms in the numerator must cancel with the one $a(t)$ term in the
denominator. That occurs precisely when $\ddot{a}(t)$ is proportional to $a(t)$.

$$\ddot{a}(t) \propto a(t).$$

Solving $\ddot{a}(t) = \zeta\, a(t)$, where ζ is a proportionality constant, yields

$$a(t) = e^{t\sqrt{\zeta}}\,\alpha + e^{-t\sqrt{\zeta}}\,\beta$$

where α and β are constants of integration, not both zero.

Also, $\zeta \neq 0$ or else $a(t)$ would be constant, violating the Case 1A assumptions.

With this value of $a(t)$, the $a(t)$ terms in the numerator
cancel with the $a(t)$ term in the denominator, leaving $A(r)$ a constant:

$A(r) = K$ for some constant, K.

With this value for $A(r)$, equations R_{00} and
R_{22} are dependent, and we are free to declare R_{00} redundant.

This reduces our situation to 4 equations in the two unknowns, D_4 and D_5:

$$R_{11}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,K} + \frac{c^2\,\ddot{a}}{a\,K} + \frac{(D_4')^2}{4\,a^2\,D_4^2} + \frac{(D_5')^2}{4\,a^2\,D_5^2} - \frac{D_4''}{2\,a^2\,D_4} - \frac{D_5''}{2\,a^2\,D_5} = \Lambda$$

$$R_{22}: \quad \frac{K\,\Lambda}{\sqrt{3}\,c\,\sqrt{\kappa}} = \Lambda$$

$$R_{44}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,K} + \frac{c^2\,\ddot{a}}{a\,K} + \frac{(D_4')^2}{4\,a^2\,D_4^2} - \frac{D_4'\,D_5'}{4\,a^2\,D_4\,D_5} - \frac{D_4''}{2\,a^2\,D_4} = \Lambda$$

$$R_{55}: \quad \frac{2\,c^2\,\dot{a}^2}{a^2\,K} + \frac{c^2\,\ddot{a}}{a\,K} - \frac{D_4'\,D_5'}{4\,a^2\,D_4\,D_5} + \frac{(D_5')^2}{4\,a^2\,D_5^2} - \frac{D_5''}{2\,a^2\,D_5} = \Lambda$$

From R_{22} we have a value for K:

$$K = \sqrt{3}\,c\,\sqrt{\kappa}$$

We can solve R_{11}–R_{55} for D_5:

$$R_{11}-R_{55}: \quad \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} + \frac{\mathcal{D}_4'\,\mathcal{D}_5'}{4\,a^2\,\mathcal{D}_4\,\mathcal{D}_5} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} = 0$$

$$\mathcal{D}_5(r) = \frac{\gamma\,(\mathcal{D}_4')^2}{\mathcal{D}_4}$$ where γ is a constant of integration.

Since $g_{55}=-a(t)^2\mathcal{D}_5(r)p_5{}^2$, we can roll the positive part of γ into the radius p_5:

$$\mathcal{D}_5(r) = \pm\frac{(\mathcal{D}_4')^2}{\mathcal{D}_4}.$$

With this formula, R_{11} and R_{55} are dependent equations and we are free to designate R_{11} as redundant. Then the remaining equations become:

$$R_{44}: \quad \frac{2\,c\,\dot{a}^2}{\sqrt{3}\,a^2\,\sqrt{\kappa}} + \frac{c\,\ddot{a}}{\sqrt{3}\,a\,\sqrt{\kappa}} + \frac{(\mathcal{D}_4')^2}{2\,a^2\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{a^2\,\mathcal{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{2\,c\,\dot{a}^2}{\sqrt{3}\,a^2\,\sqrt{\kappa}} + \frac{c\,\ddot{a}}{\sqrt{3}\,a\,\sqrt{\kappa}} - \frac{\mathcal{D}_4'''}{a^2\,\mathcal{D}_4'} - \frac{(\mathcal{D}_4')^2}{2\,a^2\,\mathcal{D}_4^2} + \frac{\mathcal{D}_4''}{a^2\,\mathcal{D}_4} = \Lambda$$

R_{44} can be used to solve for $\mathcal{D}_4$:

$$\mathcal{D}_4(r) = \chi\,\mathrm{Cos}\left[\frac{(r-\omega)\,\sqrt{3\,a^2\,\sqrt{\kappa}\,\Lambda - 2\,\sqrt{3}\,c\,\dot{a}^2 - \sqrt{3}\,a\,c\,\ddot{a}}}{\sqrt{6}\,\kappa^{1/4}}\right]^2$$

where ω and χ are constants of integration.

Since $g_{44}=-a(t)^2\mathcal{D}_4(r)p_4{}^2$, we can roll the positive part of χ into the radius p_4:

$$\mathcal{D}_4(r) = \pm\mathrm{Cos}\left[\frac{(r-\omega)\,\sqrt{3\,a^2\,\sqrt{\kappa}\,\Lambda - 2\,\sqrt{3}\,c\,\dot{a}^2 - \sqrt{3}\,a\,c\,\ddot{a}}}{\sqrt{6}\,\kappa^{1/4}}\right]^2.$$

In order that $\mathcal{D}_4$ be strictly a function of r, the expression in the square root must be a function of r, which means it must reduce to a constant since it is a function only of t.

Let's call the expression in the square root s(t):

$$s(t) = 3\,\sqrt{\kappa}\,\Lambda\,a[t]^2 - 2\,\sqrt{3}\,c\,\dot{a}[t]^2 - \sqrt{3}\,c\,a[t]\,\ddot{a}[t]$$

Recalling that

$$a(t) = e^{t\sqrt{\zeta}}\,\alpha + e^{-t\sqrt{\zeta}}\,\beta$$ where at least one of α or β is non-zero,

then

$$s(t) = e^{2t\sqrt{\zeta}}\,\alpha^2\left(-3\,\sqrt{3}\,c\,\zeta + 3\,\sqrt{\kappa}\,\Lambda\right) + e^{-2t\sqrt{\zeta}}\,\beta^2\left(-3\,\sqrt{3}\,c\,\zeta + 3\,\sqrt{\kappa}\,\Lambda\right) + \alpha\,\beta\left(2\,\sqrt{3}\,c\,\zeta + 6\,\sqrt{\kappa}\,\Lambda\right)$$

Claim: in order for $s(t)$ to reduce to a constant, $\zeta = \frac{\sqrt{\kappa}\,\Lambda}{\sqrt{3}\,c}$:

To reduce to a constant, the $e^{2t\sqrt{\zeta}}$ and $e^{-2t\sqrt{\zeta}}$ terms must cancel out.

But, the terms can't be combined.

However, if $\zeta = \frac{\sqrt{\kappa}\,\Lambda}{\sqrt{3}\,c}$, then $-3\sqrt{3}\,c\zeta+3\sqrt{\kappa}\,\Lambda = 0$, which zeroes both terms.

With this value for ζ, the expressions for $s(t)$ and $D_4(r)$ reduce to

$$s(t) = 8\,\alpha\,\beta\,\sqrt{\kappa}\,\Lambda$$

$$D_4(r) = \pm\mathrm{Cos}^2\left[\frac{r-\omega}{\sqrt{6\sqrt{\kappa}}}\sqrt{s[t]}\right] = \pm\mathrm{Cos}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]^2$$

Plugging this formula into the two remaining equations yields:

R_{44}: $\Lambda = \Lambda$

R_{55}: $\Lambda = \Lambda$

We can simplify the expression for $a(t)$ and
also make it match the $a(t)$ expression in the 5D case by setting

$$\kappa = \frac{3\,c^2\,k^4}{\Lambda^2}$$

$$\Rightarrow \quad a(t) = e^{k\,t}\,\alpha + e^{-k\,t}\,\beta$$

Plugging in $D_4(r)$, the formula for $D_5(r)$ becomes

$$D_5(r) = \pm\frac{16}{3}\,\alpha\,\beta\,\Lambda\,\mathrm{Sin}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]^2.$$

This can be simplified by rolling $\frac{16}{3}\alpha\,\beta\,\Lambda$ into p_5:

$$D_5(r) = \pm\mathrm{Sin}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]^2.$$

All of the equations have been satisfied,
so we have an exact solution to the $\Lambda\neq 0$ field equations:

$a(t) = e^{k\,t}\,\alpha + e^{-k\,t}\,\beta$ where k, α and β are free constants,

$A(t,r) = \frac{3\,c^2\,k^2}{\Lambda}$ where $\Lambda\neq 0$,

$B(t,r) = a(t)^2$

$$\mathbb{C}(t,r) = \frac{1}{r^2 \Lambda},$$

$$\mathbb{D}_4(t,r) = \pm a(t)^2 \mathcal{D}_4(r) = \pm a[t]^2 \operatorname{Cos}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2$$

where ω is a free constant,

$$\mathbb{D}_5(t,r) = \pm a(t)^2 \mathcal{D}_5(r) = \pm a[t]^2 \operatorname{Sin}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2,$$

and line element

$$ds^2 = \frac{3c^2k^2}{\Lambda} dt^2 - a[t]^2 dr^2 - \frac{1}{\Lambda} d\theta^2 - \frac{\operatorname{Sin}[\theta]^2}{\Lambda} d\phi^2 - a[t]^2 \operatorname{Cos}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2 p_4^2\, d\psi_4^2 - a[t]^2 \operatorname{Sin}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2 p_5^2\, d\psi_5^2$$

We observe that this is a new solution; it is neither a generalization nor a special case of the Case d solution:

- The free $a(t()$ parameter of Case d is a generalization of $a(t) = e^{kt}\alpha + e^{-kt}\beta$, the Case d_1$\beta$ parameter.
- However, this means that the Case d $A(t,r)$ parameter does not generalize the Case d_1β parameter:

  When we plug $a(t)$ into $A(t,r) = \frac{3c^2 \Lambda \dot{a}[t]^2}{3c^2\kappa + \Lambda^2 a[t]}$ we get

  $A(t,r) = \frac{3c^2 e^{-kt} k^2 \left(-e^{2kt}\alpha + \beta\right)^2 \Lambda^3}{9c^4 e^{kt} k^4 + \left(e^{2kt}\alpha + \beta\right)\Lambda^4}$, which is not a generalization of

  $A(t,r) = \frac{3c^2k^2}{\Lambda}$.

d_II: $B(t,r)=a(t)^2$, $\mathbb{C}(t,r)=\mathbb{C}(r)$, $A(t,r)=c^2$

SOLUTION 1

One solution to this case, which we label 'Solution 1', comes from plugging $A(t,r)=c^2$ and the $a(t)$ formula into the solution, above, to Case d:

$$a(t) = \pm\frac{\sqrt{3}\, c\sqrt{\kappa}\operatorname{Sinh}\left[\sqrt{\frac{\Lambda}{3}}(t-\upsilon)\right]}{\Lambda}$$ where κ and υ are free constants,

$$A(t,r) = c^2,$$

$$B(t,r) = a(t)^2,$$

$$\mathbb{C}(t,r) = \frac{1}{r^2\Lambda}$$ where $\Lambda \neq 0$,

$$\mathbb{D}_4(t,r) = \pm a(t)^2 \mathbb{D}_4(r) = \pm a[t]^2 \operatorname{Cosh}\left[\frac{c\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2$$

where ω id a free constant,

$$\mathbb{D}_5(t,r) = \pm a(t)^2 \mathbb{D}_5(r) = \pm a[t]^2 \operatorname{Sinh}\left[\frac{c\sqrt{\kappa}\,(r-\omega)}{\sqrt{\Lambda}}\right]^2,$$

This solution, of course, is a special case of the Case a solution. However, as we see below, it is possible to get a different solution when we set $A(t,r) = c^2$ before we solve the ODEs rather than afterwards. In fact, in this case we find two new solutions.

Other solutions:

The field equations become

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}}{a\,r} - \frac{\dot{a}\,\mathbb{C}'}{a\,\mathbb{C}} = 0$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} + \frac{(\mathbb{C}')^2}{2\,a^2\,\mathbb{C}^2} + \frac{(\mathbb{D}_4')^2}{4\,a^2\,\mathbb{D}_4^2} + \frac{(\mathbb{D}_5')^2}{4\,a^2\,\mathbb{D}_5^2} - \frac{\mathbb{C}''}{a^2\,\mathbb{C}} - \frac{\mathbb{D}_4''}{2\,a^2\,\mathbb{D}_4} - \frac{\mathbb{D}_5''}{2\,a^2\,\mathbb{D}_5} = \Lambda$$

$$R_{22}: \quad -\frac{1}{a^2\,r^2} + \frac{1}{r^2\,\mathbb{C}} - \frac{2\,\mathbb{C}'}{a^2\,r\,\mathbb{C}} - \frac{\mathbb{D}_4'}{2\,a^2\,r\,\mathbb{D}_4} - \frac{\mathbb{C}'\,\mathbb{D}_4'}{4\,a^2\,\mathbb{C}\,\mathbb{D}_4} - \frac{\mathbb{D}_5'}{2\,a^2\,r\,\mathbb{D}_5} - \frac{\mathbb{C}'\,\mathbb{D}_5'}{4\,a^2\,\mathbb{C}\,\mathbb{D}_5} - \frac{\mathbb{C}''}{2\,a^2\,\mathbb{C}} = \Lambda$$

$$R_{44}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{\mathbb{D}_4'}{a^2\,r\,\mathbb{D}_4} - \frac{\mathbb{C}'\,\mathbb{D}_4'}{2\,a^2\,\mathbb{C}\,\mathbb{D}_4} + \frac{(\mathbb{D}_4')^2}{4\,a^2\,\mathbb{D}_4^2} - \frac{\mathbb{D}_4'\,\mathbb{D}_5'}{4\,a^2\,\mathbb{D}_4\,\mathbb{D}_5} - \frac{\mathbb{D}_4''}{2\,a^2\,\mathbb{D}_4} = \Lambda$$

$$R_{55}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{\mathbb{D}_5'}{a^2\,r\,\mathbb{D}_5} - \frac{\mathbb{C}'\,\mathbb{D}_5'}{2\,a^2\,\mathbb{C}\,\mathbb{D}_5} - \frac{\mathbb{D}_4'\,\mathbb{D}_5'}{4\,a^2\,\mathbb{D}_4\,\mathbb{D}_5} + \frac{(\mathbb{D}_5')^2}{4\,a^2\,\mathbb{D}_5^2} - \frac{\mathbb{D}_5''}{2\,a^2\,\mathbb{D}_5} = \Lambda$$

We can simplify R_{01} by factoring out $\frac{\dot{a}}{a}$:

$$R_{01}: \quad \frac{2}{r} + \frac{\mathbb{C}'}{\mathbb{C}} = 0$$

Solving for $\mathbb{C}$ yields:

$$\mathbb{C}[r] = \frac{K}{r^2}$$ where K is a constant of integration.

This eliminates R_{01}, and the remaining equations become

$$R_{00}: \quad \frac{3\,\ddot{a}}{a} = \Lambda$$

$$R_{11}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{(\mathbb{D}_4')^2}{4\,a^2\,\mathbb{D}_4^2} + \frac{(\mathbb{D}_5')^2}{4\,a^2\,\mathbb{D}_5^2} - \frac{\mathbb{D}_4''}{2\,a^2\,\mathbb{D}_4} - \frac{\mathbb{D}_5''}{2\,a^2\,\mathbb{D}_5} = \Lambda$$

R_{22}: $\frac{1}{K} = \Lambda$

R_{44}: $\frac{2\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{(D_4')^2}{4a^2 D_4^2} - \frac{D_4' D_5'}{4a^2 D_4 D_5} - \frac{D_4''}{2a^2 D_4} = \Lambda$

R_{55}: $\frac{2\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} - \frac{D_4' D_5'}{4a^2 D_4 D_5} + \frac{(D_5')^2}{4a^2 D_5^2} - \frac{D_5''}{2a^2 D_5} = \Lambda$

From R_{22} we get

$$\Lambda \neq 0$$

and

$$K = \frac{1}{\Lambda}.$$

Moreover, R_{00} shows that $a(t)$ is not generic for this solution:

$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\beta$ where α and β are constants of integration.

For the sake of keeping the equations more manageable, we work with $a(t)$ for now rather than this formula.

R_{11}-R_{44}-R_{55} can be solved for D_5:

R_{11}-R_{44}-R_{55}: $-\frac{2\dot{a}^2}{a^2} - \frac{\ddot{a}}{a} + \frac{D_4' D_5'}{2a^2 D_4 D_5} = -\Lambda$

$D_5(r) = \xi\, e^{-2(a^2\Lambda - 2\dot{a}^2 - a\ddot{a})\int_1^r \frac{2 D_4}{D_4'} d\rho}$ where ξ is a constant of integration.

We can roll the positive part of ξ into the radius p_5:

$$D_5(r) = \pm e^{-2(a^2\Lambda - 2\dot{a}^2 - a\ddot{a})\int_1^r \frac{2 D_4}{D_4'} d\rho}$$

With this formula, the remaining equations, R_{11}, R_{44}, and R_{55}, become unwieldy. Nonetheless, R_{44}-R_{55} is complex but manageable. It can be solved for the remaining unknown, D_4, generating three solutions.

$$R_{44}-R_{55}: \quad -\Lambda - \frac{1}{4}\left(\int_1^r \frac{2\left(5\,\dot{a}\,\ddot{a} + a\left(-2\,\Lambda\,\dot{a} + \dddot{a}\right)\right)\mathcal{D}_4}{\mathcal{D}_4'[\rho]}\,d\rho\right)^2 -$$

$$\frac{1}{2}\int_1^r \frac{2\left(-2\,\Lambda\,\dot{a}^2 + 5\,\ddot{a}^2 + 6\,\dot{a}\,\ddot{a} + a\left(-2\,\Lambda\,\ddot{a} + \text{OverDot}[a,4]\right)\right)\mathcal{D}_4}{\mathcal{D}_4'[\rho]}\,d\rho -$$

$$\frac{3\left(\int_1^r \frac{2\left(5\,\dot{a}\,\ddot{a}+a\left(-2\,\Lambda\,\dot{a}+\dddot{a}\right)\right)\mathcal{D}_4}{\mathcal{D}_4'[\rho]}\,d\rho\right)\dot{a}}{2\,a} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} + \frac{a^2\,\Lambda^2\,\mathcal{D}_4^2}{(\mathcal{D}_4')^2} - \frac{4\,\Lambda\,\dot{a}^2\,\mathcal{D}_4^2}{(\mathcal{D}_4')^2} + \frac{4\,\dot{a}^4\,\mathcal{D}_4^2}{a^2\,(\mathcal{D}_4')^2} - \frac{2\,a\,\Lambda\,\ddot{a}\,\mathcal{D}_4^2}{(\mathcal{D}_4')^2} +$$

$$\frac{4\,\dot{a}^2\,\ddot{a}\,\mathcal{D}_4^2}{a\,(\mathcal{D}_4')^2} + \frac{\ddot{a}^2\,\mathcal{D}_4^2}{(\mathcal{D}_4')^2} + \frac{(\mathcal{D}_4')^2}{4\,a^2\,\mathcal{D}_4^2} - \frac{\mathcal{D}_4''}{2\,a^2\,\mathcal{D}_4} + \frac{\Lambda\,\mathcal{D}_4\,\mathcal{D}_4''}{(\mathcal{D}_4')^2} - \frac{2\,\dot{a}^2\,\mathcal{D}_4\,\mathcal{D}_4''}{a^2\,(\mathcal{D}_4')^2} - \frac{\ddot{a}\,\mathcal{D}_4\,\mathcal{D}_4''}{a\,(\mathcal{D}_4')^2} = 0$$

Using first two solutions of R_{44}-R_{55}:

The first two solutions are

$\mathcal{D}_4(r) = e^{\pm\frac{4\,r\,\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}}{\sqrt{3}}}\,\Xi$ where Ξ is a constant of integration.

When we roll Ξ into p_4, we double the number of solutions:

$$\Rightarrow \quad \mathcal{D}_4(r) = \pm e^{\pm\frac{4\,r\,\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}}{\sqrt{3}}}.$$

These solutions generate the same result, so we proceed using just:

$$\mathcal{D}_4 = e^{\frac{4\,r\,\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}}{\sqrt{3}}}$$

With this formula for $\mathcal{D}_4$, we can evaluate the integral in the formula for $\mathcal{D}_5$:

$$\mathcal{D}_5(r) = \pm e^{-\frac{4\,(-1+r)\,\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}}{\sqrt{3}}}.$$

Plugging $\mathcal{D}_4$ back into the field equations, we see that R_{55} is redundant, the same as R_{44}:

$$R_{11}: \quad -\frac{8\,\alpha\,\beta\,\Lambda}{3\,a^2} + \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

$$R_{44}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

$$R_{55}: \quad \frac{2\,\dot{a}^2}{a^2} + \frac{\ddot{a}}{a} = \Lambda$$

Also R_{11}-R_{44}: $\quad -\frac{8\,\alpha\,\beta\,\Lambda}{3\,a[t]^2} = 0$

For this equation to be satisfied we must have either $\alpha=0$ or $\beta=0$.

($\Lambda=0$ also satisfies this equation, but we showed above that $\Lambda\neq 0$.)

This makes R_{11} redundant, the same as R_{44}.

Plugging in the formula for a(t), R_{44} becomes

$$R_{44}: \quad \frac{\left(3\, e^{\frac{4 t \sqrt{\Lambda}}{\sqrt{3}}} \alpha^2 - 2\, e^{\frac{2 t \sqrt{\Lambda}}{\sqrt{3}}} \alpha \beta + 3 \beta^2\right) \Lambda}{3 \left(e^{\frac{2 t \sqrt{\Lambda}}{\sqrt{3}}} \alpha + \beta\right)^2} = \Lambda,$$

which is the identity $\Lambda=\Lambda$ when either $\alpha=0$ or $\beta=0$.

Also, because one of α or β is zero, $\mathbb{D}_4(r)$ and $\mathbb{D}_5(r)$ become constants, the positive part of which can be rolled into their respective radii, p_4 and p_5:

$$\mathbb{D}_4(r) = \pm 1,$$

$$\mathbb{D}_5(r) = \pm 1.$$

We can next simplify the expression for a(t):

$$a(t) = e^{\pm \frac{t \sqrt{\Lambda}}{\sqrt{3}}} \alpha,$$

SOLUTION 2

All of the equations are satisfied,
so we have an exact solution to the $\Lambda \neq 0$ field equations:

$$a(t) = e^{\pm \frac{t \sqrt{\Lambda}}{\sqrt{3}}} \alpha \quad \text{where } \alpha \text{ is a free constant,}$$

$$A(t,r) = c^2,$$

$$B(t,r) = a(t)^2$$

$$\mathbb{C}(t,r) = \frac{1}{r^2 \Lambda},$$

$$\mathbb{D}_4(t,r) = \pm a(t)^2 \mathbb{D}_4(r) = \pm a[t]^2,$$

$$\mathbb{D}_5(t,r) = \pm a(t)^2 \mathbb{D}_5(r) = \pm a[t]^2,$$

and line element

$$ds^2 = c^2\, dt^2 - a[t]^2\, dr^2 - \frac{1}{\Lambda}\, d\theta^2 - \frac{\mathrm{Sin}[\theta]^2}{\Lambda}\, d\phi^2 - a[t]^2 p_4^2\, d\psi_4^2 - a[t]^2 p_5^2\, d\psi_5^2$$

This solution is fairly trivial, but it is new. It is easy to show it is neither a generalization nor a special case of Solution 1:

In order for $\mathbb{D}_4(t,r)$ in Solution 1 to equal $a(t)^2$ we would have to set $\kappa=0$, but then a(t) would be zero, not allowed.

Thus, $\mathbb{D}_4$ in Solution 2 is not a special case of $\mathbb{D}_4$ in Solution 1.

Clearly $\mathbb{D}_4$ in Solution 1 is also not a special case of $\mathbb{D}_4(t,r)=a(t)^2$

in Solution 2.

It is also neither a generalization nor a special case of Case d:

The free parameter a(t) of Case a is a generalization of

$a(t) = \alpha e^{\pm\sqrt{\frac{\Lambda}{3}}\,t}$ from Solution 2.

$$\Rightarrow \quad \text{Case d: } A(t,r) = \frac{c^2\, e^{\frac{2t\sqrt{\Lambda}}{\sqrt{3}}}\, \alpha^2 \Lambda^2}{3\, c^2 \kappa + e^{\frac{2t\sqrt{\Lambda}}{\sqrt{3}}}\, \alpha^2 \Lambda^2} = c^2 \text{ when } \kappa=0 \quad \checkmark$$

$$\kappa=0 \Rightarrow D_4(r) = a(t)^2\, \text{Cosh}^2[0] = a(t)^2 \quad \checkmark$$

$$\text{but} \Rightarrow D_5(r) = a(t)^2\, \text{Sinh}^2[0] = 0 \quad \sharp$$

Similarly, it is neither a generalization nor a special case of Case dIβ:

Either α or β must be zero in order for the solution 2 a(t) to be a special case of Case dIβ. But that means that $D_5(r) = 0$. $\sharp$

Using third solution of R_{44}-R_{55}:

The third solution for $D_4(r)$ from R_{44}-R_{55} is:

$$D_4(r) = \Omega\, \text{Cos}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]^2 \quad \text{where } \Omega \text{ and } \omega \text{ are constants of integration.}$$

$$= \pm\text{Cos}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]^2 \quad \text{where } \Omega \text{ has been rolled into } p_4.$$

As with Solution 2, the integral in $D_5(r)$ can be evaluated when $D_4(r)$ is plugged in:

$$D_5(r) = \pm e^{2\left(\text{Log}\left[-\text{Sin}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]\right]-\text{Log}\left[-\text{Sin}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(-1+\omega)}{\sqrt{3}}\right]\right]\right)}$$

$$= \pm\text{Csc}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(-1+\omega)}{\sqrt{3}}\right]^2 \text{Sin}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]^2.$$

The cosecant term is a constant that can be either positive or negative, and thus can be rolled into the radius p_5:

$$D_5(r) = \pm\text{Sin}\left[\frac{2\sqrt{\alpha}\,\sqrt{\beta}\,\sqrt{\Lambda}\,(r-\omega)}{\sqrt{3}}\right]^2$$

The field equations are now all satisfied:

R_{11}: $\Lambda = \Lambda$

R_{44}: $\Lambda = \Lambda$

R_{55}: $\Lambda = \Lambda$

SOLUTION 3

This exact solution to the $\Lambda \neq 0$ field equations is

$$a(t) = e^{\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\alpha + e^{-\frac{t\sqrt{\Lambda}}{\sqrt{3}}}\beta \quad \text{where } \alpha \text{ and } \beta \text{ are free constants,}$$

$$A(t,r) = c^2,$$

$$B(t,r) = a(t)^2,$$

$$\mathbb{C}(t,r) = \frac{1}{r^2\Lambda},$$

$$\mathbb{D}_4(t,r) = \pm a(t)^2 \mathbb{D}_4(r) = \pm a[t]^2 \operatorname{Cos}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2$$

where ω is a free constant,

$$\mathbb{D}_5(t,r) = \pm a(t)^2 \mathbb{D}_5(r) = \pm a[t]^2 \operatorname{Sin}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2,$$

and line element

$$ds^2 = c^2\,dt^2 - a[t]^2\,dr^2 - \frac{1}{\Lambda}\,d\theta^2 - \frac{\operatorname{Sin}[\theta]^2}{\Lambda}\,d\phi^2 - a[t]^2 \operatorname{Cos}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2 p_4^2\,d\psi_4^2$$
$$- a[t]^2 \operatorname{Sin}\left[\frac{2\sqrt{\alpha}\sqrt{\beta}\sqrt{\Lambda}(r-\omega)}{\sqrt{3}}\right]^2 p_5^2\,d\psi_5^2$$

This solution is not new. It is a special case of Case dIβ: one only has to set $k = \sqrt{\frac{\Lambda}{3}}$.